CASE STUDY

China-Attributed Threat Actors: The Public Record

A TeamStarWolf case study. Published 3 November 2026; every source checked as of that date.

Executive summary

When defenders, journalists and governments talk about "Chinese hackers", they are describing dozens of distinct clusters with different sponsors, different missions and very different evidence behind them. Some are military units named in a US indictment more than a decade ago. Some are Ministry of State Security bureaus working through front companies. Some are private contractors that sell stolen mailboxes by the inbox, and some are botnet operators whose customers are other hacking teams. This study pulls the public record on the main ones into one place: who the governments of the United States, the United Kingdom, Japan, the Czech Republic, Australia and their partners have named, on what date, with what confidence, and what the advisories say the actors actually did.

Four patterns stand out. First, the sponsor picture is now unusually specific. US charging documents, Treasury designations and joint advisories name PLA Unit 61398 and the PLA 54th Research Institute, five provincial or municipal State Security bureaus (Jiangsu, Tianjin, Hainan, Hubei, Shanghai), the Ministry of Public Security, and twelve companies that hacked, brokered data or ran infrastructure for them. Second, the tradecraft has moved to the network edge. The 2021 advisories still describe spearphishing and watering holes; the 2023 to 2025 advisories on Volt Typhoon and Salt Typhoon describe routers, VPN concentrators, firewalls and living off the land with built-in tools, and every one of the 43 initial-access CVEs named across five joint advisories is in CISA's Known Exploited Vulnerabilities catalog. Third, the purpose has widened from espionage to pre-positioning: in February 2024 the US agencies that wrote AA24-038A (CISA, NSA, FBI and their US co-authors) assessed with high confidence that Volt Typhoon was positioning inside US critical infrastructure IT networks to disrupt operational technology in a crisis, with some footholds held for at least five years; partners in Australia, Canada, the United Kingdom and New Zealand co-released the advisory. Fourth, healthcare and life sciences appear throughout: from the 2014 Anthem and Community Health Systems breaches to Ebola and COVID-19 research theft, to a 2023 to 2025 campaign against medical research servers and an October 2026 advisory that names the Healthcare and Public Health sector among the targets.

Evidence grade: the indictments, sanctions, government attributions and joint advisories are **Confirmed** from the issuing agency's own page. Vendor attributions (Microsoft, Mandiant and Google Threat Intelligence Group, BAE Systems) are **Confirmed** where the vendor's own report was opened, and carry the vendor's own confidence words. The Chinese government's counter-narrative, two press accounts of an extradition and an arrest, and one press account of a 2026 seizure are **Reported**. Where this study draws its own conclusion, it is labelled **Inferred**.

Key facts at a glance

Field	Value
ATT&CK groups with a China attribution	47 of 178 Enterprise groups in ATT&CK v20.0, as the library's data holds it; 46 once Scarlet Mimic (no government link) is set aside for the technique analysis [1][3]
ATT&CK for ICS and Mobile	None of the 47 appears in the ICS or Mobile group files [7]
Distinct Enterprise techniques across the 46	320; the most shared is T1105 Ingress Tool Transfer (23 groups) [3]
Largest technique footprints	Mustang Panda G0129 (85 techniques, 23 software), APT41 G0096 (82, 32), Volt Typhoon G1017 (81, 17) [8][9][10]
Sponsors named in government records	PLA Unit 61398 and the PLA 54th Research Institute; MSS State Security bureaus in Jiangsu, Tianjin, Hainan, Hubei and Shanghai; the Ministry of Public Security [18][20][21][24][28][35][42][52]
US charging announcements covered	13, from 19 May 2014 (Unit 61398) to 8 July 2025 (Xu Zewei arrest) [18][19][20][21][23][24][26][27][28][34][42][47][52]
Treasury (OFAC) designations covered	5 actions, 25 March 2024 to 5 March 2025 [35][43][44][46][49]
Court-authorized disruptions	KV Botnet (announced 31 January 2024) and the Integrity Technology Group botnet (18 September 2024); also Microsoft's court-ordered NICKEL website seizure (December 2021) [32][38][61]
Largest healthcare data theft in the record	About 78.8 million people's records from Anthem, 2014 to 2015 [23][89]
Exploited CVEs	43 distinct initial-access CVEs across AA24-038A, AA25-239A, AA24-190A, AA22-158A and AA22-279A, all in CISA KEV; 29 marked known for ransomware use [100]
Longest dwell in the record	At least five years in some Volt Typhoon victims [33]; a 393-day average across BRICKSTORM intrusions [70]
Government time-to-mitigate benchmark	Patch or mitigate internet-facing devices within 48 hours (AA24-190A) [37]

Key findings

- **The sponsor is usually named, and it is usually the MSS.** Of the government attributions in this study, the Ministry of State Security or one of its bureaus is named for APT10, APT40, APT31, HAFNIUM-related activity, the Jiangsu turbofan case and the Salt Typhoon companies; the PLA is named for APT1, the Equifax intrusion and the AA25-239A companies; the Ministry of Public Security is named for Sichuan Silence and i-Soon (Confirmed [18][20][21][24][28][35][42][47][52][53]).
- **Contractors carry much of the load.** Huaying Haitai, Boyusec, Hainan Xiandun, Wuhan Xiaoruizhi, Shanghai Powerock, Sichuan Silence, i-Soon, Shanghai Heiying, Integrity Technology Group, Sichuan Juxinhe, Beijing Huanyu Tianqiong and Sichuan Zhixin Ruijie all appear in US or partner records, and the FBI says the MSS and MPS "weaponize" InfoSec companies (Confirmed [19][21][28][35][42][47][48][49][44][53]).
- **Edge devices are the front door.** Exploit Public-Facing Application (T1190) is named in all six advisories whose technique lists this study verified, and it holds at 10 of the 15 groups this study profiles most closely while phishing falls to 6 of 15 in that set (Confirmed [29][31][33][37][97][53][3]; the reading of the shift is Inferred).
- **Pre-positioning is the new mission.** The US agencies that wrote AA24-038A (CISA, NSA, FBI and their US co-authors) assessed with high confidence that Volt Typhoon is pre-positioning to disrupt OT in Communications, Energy, Transportation and Water, in an advisory co-released with partners in Australia, Canada, the UK and New Zealand (Confirmed [33]). Microsoft's earlier view was moderate confidence on intent (Confirmed [62]).
- **Healthcare is a recurring target, mostly for intellectual property and identity data.** The record runs from Anthem and Community Health Systems (2014) through Ebola research theft and the targeting of HIV and COVID-19 research, to UNC6508's medical research campaign (2023 to 2025) and the Healthcare and Public Health sector named in AA26-281A (October 2026) (Confirmed [23][56][87][26][78][80]).
- **Names do not line up, and the disagreements are on the record.** The FBI lists Silk Typhoon as a name for the APT27 actors Yin Kecheng and Zhou Shuai; ATT&CK and Microsoft file Silk Typhoon under HAFNIUM; Google says UNC5221 is not Silk Typhoon while Microsoft lists under Silk Typhoon CVE-2025-0282, which Mandiant attributes to UNC5221 (Confirmed [48][1][68][69][70]).
- **Some of the biggest names have no ATT&CK object.** Flax Typhoon, Storm-0558, UNC5221 (now ARMORY CASTLE), UNC2814, UNC6508 and Storm-2603 have no group page in the library; Microsoft's Linen Typhoon and Violet Typhoon are filed as aliases of G0027 Threat Group-3390 and G0128 ZIRCONIUM (Confirmed [1][6]).
- **Covert networks of compromised devices are now the default infrastructure.** Sixteen agencies say such networks are believed used by the majority of China-linked actors and are built by Chinese information security companies (Confirmed [74]).

Top recommendations

1. **Run an edge-first patch queue.** Put internet-facing VPN, firewall, router and web application CVEs from the KEV catalog at the head of the queue, and measure time-to-mitigate against the 48-hour window in AA24-190A (Confirmed [37][100]; the program rule is Inferred).
2. **Treat network devices as monitored hosts.** Baseline running configurations, verify firmware and image hashes, put management on an isolated VRF or out-of-band network, and log AAA command accounting

centrally, as AA25-239A asks (Confirmed [53]).

3. **Detect living off the land on identity infrastructure.** Enable process creation with command lines and deep PowerShell logging, alert on NTDS.dit copying through ESENT events and on every Security log clear, and review VPN and RDP logons for impossible travel (Confirmed [31][33]).
4. **Retire end-of-life SOHO and edge equipment.** The KV Botnet ran mostly on end-of-life Cisco and NetGear routers, and the Integrity Tech botnet passed 260,000 devices (Confirmed [32][39]).
5. **For hospitals and research institutions: inventory research applications and third-party access.** REDCap servers were the path in UNC6508's 2023 to 2025 campaign (Confirmed [78]), and Microsoft names healthcare among the sectors Silk Typhoon targets through stolen IT provider, RMM and MSP credentials (Confirmed [69]); audit mail content compliance rules, which UNC6508 abused to BCC-forward mail (Confirmed [78]); review transport rules too (Inferred).
6. **Read the attribution words, not just the name.** Record each source's confidence language ("almost certain", "highly likely", "moderate confidence", "suspected") alongside the actor name in threat intelligence tickets, because the sources disagree in degree (Confirmed [30][62][70][73]; the practice is Inferred).

Scope and method

This study covers threat actors that governments or the ATT&CK knowledge base attribute to the People's Republic of China, from the earliest activity the record dates (2006) to October 2026. It draws on US Department of Justice releases, charging documents and an affidavit; FBI wanted notices and public service announcements; Treasury OFAC press releases; CISA, NSA and FBI joint advisories and their partner co-seals; UK government and NCSC statements; the Czech government and NUKIB; Japan's National Police Agency; the Congressional Research Service; HHS's Health Sector Cybersecurity Coordination Center (HC3) briefings; vendor primary reports from Microsoft, Mandiant, Google Threat Intelligence Group (GTIG) and BAE Systems; a Community Health Systems SEC filing; the live MITRE ATT&CK group pages and the library's own ATT&CK v20.0 data files. Every source was opened on 3 November 2026. Quotations are kept to a few words.

Grade	Meaning
Confirmed	A primary source was opened: the government agency, court or regulator, the vendor that observed the activity, the victim's own filing, or the MITRE record itself
Reported	Known only through a secondary source, such as press coverage or state media, whose original was not opened
Demonstrated	A public reproduction exists. No item here is graded Demonstrated: this study describes actors, not vulnerabilities, and no exploit repository was opened
Inferred	This study's own reasoning from the cited facts, labelled where it appears

Which groups count as China-attributed. The inventory in Part 1 includes an ATT&CK group only when its ATT&CK description attributes it to China, the PRC, a PRC body or a Chinese-speaking operator (Confirmed [1]). Groups that ATT&CK names China only as a target, and groups whose aliases follow Panda, Dragon or Typhoon naming but whose ATT&CK description carries no China wording, are listed separately and left out of the counts. Vendor reporting probably ties several of those to China, but that was not checked against a primary source here, so they are not described as China-attributed (Inferred [1]).

Attribution language. Each attribution is given in the source's own words: "almost certain", "highly likely", "high confidence", "moderate confidence", "suspected", "consistent with" and "linked to" are carried through. Where two sources disagree, the study says so in the text and in Cross-cutting themes.

Home studies. APT41 has its own companion study, APT41: The Public Record; this study summarizes it in one paragraph. The 2025 SharePoint exploitation is covered in The SharePoint ToolShell Mass Exploitation, and the 2021 Exchange mass exploitation in Operation Exchange Marauder.

Tradecraft level. Behavior is described at the level the advisories use: technique names and ATT&CK ids. No commands, payloads or exploit details are reproduced.

ATT&CK version. Technique ids and tactic names follow ATT&CK v20.0 as the library holds it. Since v19, ATT&CK splits the former Defense Evasion tactic into Stealth and Defense Impairment (Confirmed [101]). Older advisories cite two ids that v20.0 revokes: T1070.001 is now T1685.005 and T1562.004 is now T1686 (Confirmed [4]).

Part 1: who ATT&CK attributes to China

The inventory

The library's ATT&CK v20.0 Enterprise data (commit e85b559) holds 178 groups. Forty-seven of them have an ATT&CK description that attributes them to China, the PRC, a PRC body or a Chinese-speaking operator, and each has a generated page under /mitre/groups/ (Confirmed [1]). The library's group records carry no first-seen or last-seen field, so the "since" years below come from ATT&CK's description text; campaign dates come from campaign objects (Confirmed [7]).

Id	Name	What ATT&CK says about the sponsor or origin	Since (ATT&CK text)	Techniques	Software
G0001	Axiom	China attribution in the ATT&CK description	2008	16	8
G0002	Moafee	Guangdong Province ("Guandong" in ATT&CK)	not stated	1	1
G0004	Ke3chang	Actors operating out of China	2010	46	11
G0005	APT12	China attribution in the ATT&CK description	not stated	5	3
G0006	APT1	PLA GSD 3rd Department 2nd Bureau, Unit 61398	not stated	23	17
G0009	Deep Panda	Suspected Chinese threat group; Anthem intrusion	not stated	10	7
G0011	PittyTiger	China attribution in the ATT&CK description	not stated	2	5
G0013	APT30	China attribution in the ATT&CK description	not stated	2	5
G0018	admin@338	China attribution in the ATT&CK description	not stated	12	7
G0019	Naikon	Chengdu Military Region Second Technical Reconnaissance Bureau, MUCD 78020	2010	14	15
G0022	APT3	Attributed by researchers to the MSS	not stated	44	6
G0023	APT16	China attribution in the ATT&CK description	not stated	1	1

Id	Name	What ATT&CK says about the sponsor or origin	Since (ATT&CK text)	Techniques	Software
G0024	Putter Panda	PLA Unit 61486, 12th Bureau	not stated	4	4
G0025	APT17	China attribution in the ATT&CK description	not stated	2	1
G0027	Threat Group-3390 (APT27)	Chinese threat group known for strategic web compromises	2010	57	24
G0029	Scarlet Mimic	Motives overlap the Chinese government; not directly linked	not stated	1	4
G0039	Suckfly	China attribution in the ATT&CK description	2014	5	1
G0044	Winnti Group	China attribution in the ATT&CK description	2010	6	3
G0045	menuPass (APT10)	Members acted with the MSS Tianjin State Security Bureau; Huaying Haitai	2006	46	25
G0060	BRONZE BUTLER	China attribution in the ATT&CK description	2008	40	14
G0062	TA459	China attribution in the ATT&CK description	not stated	5	4
G0065	Leviathan (APT40)	MSS Hainan State Security Department and an affiliated front company	2009	50	17
G0066	Elderwood	China attribution in the ATT&CK description	not stated	9	9
G0073	APT19	China attribution in the ATT&CK description	not stated	21	2
G0093	GALLIUM	Likely state-sponsored	2012	31	16
G0096	APT41	Chinese state-sponsored espionage group with financially motivated operations	2012	82	32
G0098	BlackTech	China attribution in the ATT&CK description	2013	14	6
G0103	Mofang	China attribution in the ATT&CK description	May 2012	6	2
G0106	Rocke	Alleged Chinese-speaking cryptojacker	not stated	36	0
G0114	Chimera	Suspected China-based threat group	2018	59	6
G0125	HAFNIUM	Likely state-sponsored, operating out of China	January 2021	44	6
G0128	ZIRCONIUM (APT31)	China attribution in the ATT&CK description	2017	29	0
G0129	Mustang Panda	China-based cyber espionage threat actor	2012	85	23
G0131	Tonto Team	China attribution in the ATT&CK description	2009	15	6
G0136	IndigoZebra	China attribution in the ATT&CK description	2014	7	3
G0143	Aquatic Panda	China attribution in the ATT&CK description	May 2020	35	6
G1006	Earth Lusca	China attribution in the ATT&CK description	April 2019	44	9
G1007	Aoqin Dragon	China attribution in the ATT&CK description	2013	9	2
G1014	LuminousMoth	Chinese-speaking	October 2020	28	2
G1017	Volt Typhoon	PRC state-sponsored	2021	81	17

Id	Name	What ATT&CK says about the sponsor or origin	Since (ATT&CK text)	Techniques	Software
G1021	Cinnamon Tempest	China-based ransomware operator	2021	19	8
G1023	APT5	China attribution in the ATT&CK description	2007	29	13
G1034	Daggerfly	China attribution in the ATT&CK description	2012	17	6
G1042	RedEcho	PRC-related	not stated	5	1
G1045	Salt Typhoon	PRC state-backed	2019	58	1
G1048	UNC3886	China-nexus cyberespionage group	2022	49	8
G1054	MirrorFace	PRC-aligned; a subgroup under the menuPass umbrella	2019	43	16

Source for every row: the library's `groups.jsonl` and `group_profiles.jsonl` at ATT&CK v20.0, with the larger groups also checked on the live ATT&CK page (Confirmed [1][3][8][9][10][11][12][13][14][15][16][17]).

What ATT&CK says about the sponsors

ATT&CK names specific military units for three groups. APT1 (G0006) is the PLA General Staff Department's 3rd Department 2nd Bureau, Unit 61398; Putter Panda (G0024) is Unit 61486 of the 12th Bureau; Naikon (G0019) is the Chengdu Military Region Second Technical Reconnaissance Bureau, Military Unit Cover Designator 78020 (Confirmed [1]).

ATT&CK ties three groups to the Ministry of State Security: APT3 (G0022) is attributed by researchers to the MSS, Leviathan (G0065) to the Hainan State Security Department, and menuPass (G0045) to the Tianjin State Security Bureau. MirrorFace (G1054), described as PRC-aligned and active since 2019, is placed as a subgroup under the menuPass umbrella, with campaign C0060 Operation AkaiRyu (June to September 2024, Japan and Central Europe) (Confirmed [1]).

Some attributions are weak by ATT&CK's own wording and should be labelled that way wherever they are repeated: Scarlet Mimic (motives overlap the Chinese government, not directly linked), Rocke (an alleged Chinese-speaking cryptojacker), LuminousMoth (Chinese-speaking), Moafee (Guangdong Province) and RedEcho (PRC-related) (Confirmed [1]). Cinnamon Tempest (G1021) is a China-based group active since at least 2021 that deploys Babuk-derived ransomware without an affiliate model; ATT&CK says it may be motivated by intellectual property theft or espionage rather than money (Confirmed [1]).

Who is left out, and why

ATT&CK names China only as a target for Strider (G0041), APT37 (G0067), Kimsuky (G0094), Sidewinder (G0121), Higaisa (G0126) and BITTER (G1002) (Confirmed [1]). APT18 (G0026), Lotus Blossom (G0030), Tropic Trooper (G0081, which ATT&CK calls "unaffiliated") and DragonOK (G0017) carry Panda, Dragon or Typhoon style aliases but no China attribution in the ATT&CK description; Rancor (G0075), Thrip (G0076), Whitefly (G0107), BackdoorDiplomacy (G0135), ToddyCat (G1022), Velvet Ant (G1047) and Sowbug (G0054) have no country wording at all (Confirmed [1]). Vendor reporting likely ties several of these to China, but this study did not open a primary source that does, so they are not counted (Inferred [1]). APT18 returns in Part 5 because HHS's HC3 says it is believed responsible for the 2014 Community Health Systems breach (Reported [83]).

Campaigns with no group attached

ATT&CK records eleven China-linked campaigns with no group object, each with a library page (Confirmed [2]):

Campaign	Dates (ATT&CK)	What it is
C0002 Night Dragon	Nov 2009 to Feb 2011	China-linked campaign in ATT&CK
C0007 FunnyDream	Jul 2018 to Nov 2020	China-linked campaign in ATT&CK
C0012 Operation CuckooBees	Dec 2019 to May 2022	Winnti, APT41 and BARIUM-affiliated actors
C0014 Operation Wocao	Dec 2017 to Dec 2019	China-linked campaign in ATT&CK
C0029 Cutting Edge	Dec 2023 to Feb 2024	Ivanti zero-days, UNC5221 and UNC5325
C0043 Indian Critical Infrastructure Intrusions	Jan 2021 to Apr 2022	RedEcho and TAG38
C0053 FLORAHOX Activity	Jan 2019 to May 2024	China-linked campaign in ATT&CK
C0055 Quad7 Activity	Aug 2023 to Aug 2025	Storm-0940
C0058 SharePoint ToolShell Exploitation	Jul 2025	Storm-2603, TG-3390 and ZIRCONIUM; healthcare among the sectors
C0061 Operation Digital Eye	Jun to Jul 2024	China-linked campaign in ATT&CK
C0062 Anthropic AI-orchestrated Campaign	Sep 2025	GTG-1002

Part 2: the sponsor structure

The government record now maps China's offensive cyber activity onto three state bodies and a ring of companies. The table lists every sponsor and company the record in this study names, with the document that names it.

State body	Unit, bureau or company	Activity and library binding	First named in this record	Source
PLA	Unit 61398, Third Department	Five officers indicted; APT1 (G0006)	19 May 2014 (Mandiant report 19 Feb 2013)	[18][55]
PLA	54th Research Institute	Four members charged with the Equifax intrusion; no ATT&CK group bound	10 Feb 2020	[24]

State body	Unit, bureau or company	Activity and library binding	First named in this record	Source
PLA and MSS	Sichuan Juxinhe, Beijing Huanyu Tianqiong, Sichuan Zhixin Ruijie	Supply Chinese intelligence services; linked to PLA units and the MSS; Salt Typhoon cluster (G1045)	27 Aug 2025 (Juxinhe sanctioned 17 Jan 2025)	[53][46]
MSS	Jiangsu State Security Department	Officers Zha Rong and Chai Meng; turbofan engine data, 2010 to 2015	30 Oct 2018	[20]
MSS	Tianjin State Security Bureau, with Huaying Haitai	APT10, menuPass (G0045)	20 Dec 2018	[21][22]
MSS	Hainan State Security Department, with front company Hainan Xiandun	APT40, Leviathan (G0065)	19 Jul 2021 (indictment returned May 2021)	[28][29]
MSS	Hubei State Security Department, with front company Wuhan Xiaoruizhi (set up 2010)	APT31, ZIRCONIUM (G0128)	25 Mar 2024	[35]
MSS	Shanghai State Security Bureau, with Shanghai Powerock	Xu Zewei and Zhang Yu; COVID-19 research and the HAFNIUM Exchange campaign (G0125)	8 Jul 2025	[52]
MSS (alleged, for personal gain and for the MSS)	Li Xiaoyu and Dong Jiazhi	Decade-long campaign including COVID-19 research	21 Jul 2020	[26]
MSS-affiliated	Yin Kecheng	Treasury Departmental Offices compromise; APT27 (G0027)	17 Jan 2025	[46][47]
MPS	Sichuan Silence (Chengdu)	Sophos firewall zero-day, CVE-2020-12271; services to the Ministry of Public Security; core clients are PRC intelligence services	10 Dec 2024	[42][43]
MSS and MPS	i-Soon (Anxun)	Sold data to at least 43 MSS or MPS bureaus in 31 provinces; Aquatic Panda (G0143) is among its industry names; Charcoal Typhoon is filed in ATT&CK under G1006	5 Mar 2025	[47][48][96]
CCP intelligence services (Treasury: "almost certainly")	Zhou Shuai, Shanghai Heiying	Data broker since at least 2018; APT27 (G0027)	5 Mar 2025	[49]
Not stated (company with PRC government links)	Integrity Technology Group (Beijing)	Botnet of over 260,000 devices; Flax Typhoon, RedJuliett, Ethereal Panda; no ATT&CK group	18 Sep 2024	[38][39][44]
Not stated in DOJ release	Guangzhou Bo Yu (Boyusec)	Moody's Analytics, Siemens, Trimble; the DOJ release does not use the APT3 label; the link to G0022 is not checked here (Inferred)	27 Nov 2017	[19]
MSS and PLA as clients (press account of DOJ)	Nanjing Xinjiuwei Network Technology	Platforms behind a 2018 to 2026 campaign; domains seized	26 Aug 2026	[94][95]

Two points about the table. First, the MSS presence is provincial: the record names bureaus in Jiangsu, Tianjin, Hainan, Hubei and Shanghai; four of them worked through a named company (Huaying Haitai, Hainan Xiandun, Wuhan Xiaoruizhi, Shanghai Powerock), and the Jiangsu officers worked through recruited hackers and insiders (Confirmed [20][21][28][35][52]). Second, the contractor relationship runs both ways. Some companies are fronts set up by a bureau (Wuhan Xiaoruizhi in 2010, Hainan Xiandun) (Confirmed [35][28]); others are commercial firms that sell services or stolen data to many bureaus, as i-Soon did to at least 43 bureaus at USD 10,000 to 75,000 per inbox (Confirmed [47]).

The FBI's March 2025 announcement states the model plainly: the MSS and MPS use private InfoSec companies, and it lists the industry names for i-Soon, including Aquatic Panda and Charcoal Typhoon (Confirmed [48]). In April 2026, NCSC and 15 partner agencies added that covert networks of compromised devices are believed used by the majority of China-linked actors and are built by Chinese information security companies; the UK had sanctioned Integrity Technology Group and one other company in December 2025 (Confirmed [74]).

Part 3: the operational record

One row per dated event or report the verified record supports. "Actor" uses the library's group id where the source's actor maps to one; where it does not, the row says so.

Date	Event	Actor and library binding	Grade	Source
2006 onward	menuPass active since at least 2006; APT1 economic espionage since 2006 against 141 victims	G0045, G0006	Confirmed	[16][55]
Nov 2009 to Feb 2011	Night Dragon campaign	C0002	Confirmed	[2]
Dec 2009 (disclosed 12 Jan 2010)	Google discloses a targeted attack from China that stole intellectual property; at least twenty other large companies targeted; Gmail accounts of Chinese human rights activists a primary goal. No state attribution in the post	No ATT&CK binding	Confirmed	[54]
2010 to 2015	Jiangsu SSD officers and recruited hackers and insiders steal turbofan engine data via spearphishing, watering holes and domain hijacking	No ATT&CK binding	Confirmed	[20]
2011 to May 2017	Boyusec employees target Moody's Analytics, Siemens and Trimble	G0022 by vendor reporting only, not checked here (Inferred)	Confirmed (DOJ); APT3 link Inferred, no vendor source opened	[19]
2011 to 2018	Hainan Xiandun runs APT40 operations, including Ebola, MERS, HIV/AIDS, Marburg and tularemia research	G0065	Confirmed	[28][87]
19 Feb 2013	Mandiant's APT1 report ties APT1 to PLA Unit 61398 and releases over 3,000 indicators	G0006	Confirmed	[55]

Date	Event	Actor and library binding	Grade	Source
Feb 2014 to Jan 2015	Anthem: data on about 78.8 million people stolen	G0009 per ATT&CK; DOJ names no group	Confirmed	[23][89][1]
Apr and Jun 2014	Community Health Systems: non-medical identification data on about 4.5 million people copied; the company and Mandiant believe an APT group originating from China was responsible	No group in the filing; APT18 per HC3	Confirmed (filing); APT18 Reported	[56][83]
19 May 2014	W.D. Pa. indictment of five Unit 61398 officers, 31 counts, six US victims	G0006	Confirmed	[18]
2015	OPM breach; the House Oversight majority report names only "a foreign adversary"	No attribution on this primary	Confirmed	[57]
2016 to 2017	menuPass targets managed service providers	G0045	Confirmed	[16]
4 Apr 2017	BAE Systems attributes Operation Cloud Hopper against MSPs to APT10	G0045	Confirmed	[58]
27 Nov 2017	DOJ charges three Boyusec employees	G0022 by vendor reporting only, not checked here (Inferred)	Confirmed (DOJ); APT3 link Inferred	[19]
Dec 2017 to Dec 2019	Operation Wocao	C0014	Confirmed	[2]
Jul 2018 to Nov 2020	FunnyDream	C0007	Confirmed	[2]
30 Oct 2018	DOJ charges Jiangsu SSD officers Zha Rong and Chai Meng and their recruits	No ATT&CK binding	Confirmed	[20]
20 Dec 2018	APT10 indictment of Zhu Hua and Zhang Shilong; the UK and allies say APT10 acted for the MSS	G0045	Confirmed	[21][22][90]
Jan 2019 to May 2024	SPACEHOP (APT5 and Ke3chang) and FLORAHOX	C0052, C0053	Confirmed	[17][2]
Apr 2019	Suspected Chinese espionage actors target a US cancer-research health center that APT41 spearphished in 2018 and APT22 targeted earlier	Unnamed; APT41 (G0096) spearphished the same center in 2018; APT22 has no ATT&CK object	Confirmed (HC3)	[83]
9 May 2019	Anthem indictment of Fujie Wang and a John Doe; no state direction alleged	G0009 per ATT&CK	Confirmed	[23]
Dec 2019 to May 2022	Operation CuckooBees	C0012	Confirmed	[2]
2020	Sichuan Silence uses a Sophos firewall zero-day (CVE-2020-12271) against about 81,000 firewalls	No ATT&CK binding	Confirmed	[42][43]

Date	Event	Actor and library binding	Grade	Source
10 Feb 2020	Four PLA 54th Research Institute members charged over Equifax: about 145 million Americans' PII; about 34 servers in nearly 20 countries	No ATT&CK binding	Confirmed	[24]
Feb 2020 to Jun 2021	Xu Zewei and Zhang Yu, directed by the Shanghai SSB, target COVID-19 research mailboxes and take part in the HAFNIUM Exchange campaign	G0125	Confirmed	[52]
13 May 2020	FBI and CISA warn that COVID-19 healthcare, pharmaceutical and research organizations are prime PRC targets	PRC actors, no group	Confirmed	[25][84]
21 Jul 2020	Li Xiaoyu and Dong Jiazhi indicted; COVID-19 vaccine, testing and treatment networks probed	No ATT&CK binding	Confirmed	[26]
16 Sep 2020	Five PRC nationals (APT41, Barium, Winnti) and two Malaysian businessmen charged; more than 100 victims	G0096	Confirmed	[27]
Jan 2021 to Apr 2022	Indian critical infrastructure intrusions	C0043; RedEcho (G1042) named in the campaign description	Confirmed	[2]
2 Mar 2021	Microsoft attributes Exchange zero-day exploitation to HAFNIUM with high confidence	G0125	Confirmed	[59]
May 2021 to Feb 2022	APT41 campaign C0017	C0017	Confirmed	[9]
mid-2021	Volt Typhoon active (Microsoft); the Integrity Tech botnet starts	G1017; Integrity Tech has no ATT&CK object	Confirmed	[62][39]
2021 to 2022	UK Electoral Commission highly likely compromised by a China state-affiliated entity; APT31 reconnaissance of UK parliamentarians in 2021	G0128 for the reconnaissance	Confirmed	[65]
19 Jul 2021	Coordinated attribution: the US with the EU, UK and NATO names MSS-affiliated actors for the Exchange campaign; DOJ unseals the APT40 indictment; AA21-200A and AA21-200B published	G0125, G0065, G0128	Confirmed	[60][28][29][30][93]
6 Dec 2021	Microsoft's Digital Crimes Unit seizes NICKEL websites under an EDVA court order	G0004	Confirmed	[61]
Apr to Sep 2022	Leviathan Australian intrusions	C0049	Confirmed	[14]
2022 onward	Campaign against an unclassified Czech Foreign Ministry network	G0128	Confirmed	[50][51]

Date	Event	Actor and library binding	Grade	Source
7 Jun 2022	AA22-158A: PRC actors exploit telecoms and network devices	PRC actors, no group	Confirmed	[97]
6 Oct 2022	AA22-279A lists 20 CVEs exploited by PRC actors	PRC actors, no group	Confirmed	[99]
Oct 2022 to Jan 2024	KV Botnet activity	C0035	Confirmed	[10]
Jan 2023 to Jun 2024	APT41 DUST	C0040	Confirmed	[9]
15 May 2023	Storm-0558 forges tokens with an acquired MSA consumer signing key; mail at about 25 organizations	No ATT&CK object	Confirmed	[64]
24 May 2023	Microsoft and AA23-144A on Volt Typhoon living off the land	G1017	Confirmed	[62][31]
Jul 2023 to Dec 2024	RedDelta modified PlugX infection chain	C0047, G0129	Confirmed	[8]
Aug 2023 to Aug 2025	Quad7	C0055	Confirmed	[2]
16 Aug 2023	HC3 Threat Profile names the greatest China-based threats to US health	See Part 5	Confirmed	[86]
Sep 2023 to Nov 2025	UNC6508 REDCap campaign against medical research	No ATT&CK object	Confirmed	[78]
Dec 2023	Court-authorized removal of KV Botnet malware from hundreds of US SOHO routers (announced 31 Jan 2024)	G1017	Confirmed	[32]
Dec 2023 to Feb 2024	Cutting Edge, Ivanti zero-days	C0029	Confirmed	[2]
7 Feb 2024	AA24-038A: high confidence that Volt Typhoon is pre-positioning for OT disruption	G1017	Confirmed	[33]
25 Mar 2024	Seven APT31-linked nationals charged; OFAC sanctions Wuhan XRZ; UK sanctions and attributions	G0128	Confirmed	[34][35][36][65]
Jun to Aug 2024	Versa Director zero-day exploitation	C0039, G1017	Confirmed	[10]
Jun to Sep 2024	Operation AkaiRyu; Operation Digital Eye (Jun to Jul)	C0060, C0061	Confirmed	[1][2]
mid-2024	TINYSHELL-based backdoors on Juniper MX routers (Mandiant, 11 Mar 2025); RedPenguin Jul 2024 to Mar 2025	G1048, C0056	Confirmed	[71][15]
8 Jul 2024	AA24-190A, eight-country advisory on APT40	G0065	Confirmed	[37]

Date	Event	Actor and library binding	Grade	Source
Sep 2024	QTFY actors intrude at three DOE national laboratories, NIH and an HHS agency via an Ivanti Cloud Services Appliance zero-day	No ATT&CK binding	Confirmed	[94]
18 Sep 2024	Court-authorized disruption of the Integrity Tech botnet; FBI, CNMF and NSA advisory; NCSC and partners	Flax Typhoon, no ATT&CK object	Confirmed	[38][39][66]
Sep to Dec 2024	Treasury network hack tied to Yin Kecheng	G0027	Confirmed	[47][49]
early Oct 2024	Media first report Salt Typhoon in US telecoms (CRS)	G1045	Confirmed (CRS)	[67]
14 Oct 2024	China's CVERC calls Volt Typhoon a US false flag operation, seen through state media only	G1017	Reported	[63]
13 Nov 2024	FBI and CISA: PRC-affiliated actors compromised multiple telecom companies, including court-ordered law enforcement request data	No group named	Confirmed	[40]
4 Dec 2024	Enhanced Visibility and Hardening Guidance for Communications Infrastructure	No group named	Confirmed	[41]
10 Dec 2024	Guan Tianfeng indicted; OFAC designates Sichuan Silence	No ATT&CK binding	Confirmed	[42][43]
mid-Dec 2024	Ivanti Connect Secure zero-day CVE-2025-0282 exploited by UNC5221 (Mandiant, 8 Jan 2025)	No ATT&CK object	Confirmed	[68]
3 Jan 2025	OFAC designates Integrity Technology Group	Flax Typhoon, no ATT&CK object	Confirmed	[44]
8 Jan 2025	Japan's NPA and NISC on MirrorFace, suspected Chinese involvement, since about 2019	G1054	Confirmed	[45]
17 Jan 2025	OFAC sanctions Yin Kecheng and Sichuan Juxinhe	G0027, G1045	Confirmed	[46]
5 Mar 2025	12 people charged: eight i-Soon employees, two MPS officers and APT27 actors Yin Kecheng and Zhou Shuai; FBI PSA; OFAC on Zhou Shuai; Microsoft on Silk Typhoon	G0027, G0143, G0125	Confirmed	[47][48][49][69]
Mar 2025 onward	BRICKSTORM intrusions in legal, SaaS, BPO and technology; 393-day average dwell (GTIG, 24 Sep 2025)	No ATT&CK group object (UNC5221); BRICKSTORM is S9015	Confirmed	[70]
28 May 2025	Czech national attribution of the Foreign Ministry campaign to APT31	G0128	Confirmed	[50][51]
3 Jul 2025	Xu Zewei arrested in Milan (DOJ, 8 Jul 2025)	G0125	Confirmed	[52]

Date	Event	Actor and library binding	Grade	Source
from 7 Jul 2025	SharePoint exploitation by Linen Typhoon, Violet Typhoon and Storm-2603; Warlock ransomware from 18 Jul	C0058; Linen Typhoon = G0027, Violet Typhoon = G0128 by ATT&CK alias; Storm-2603 has no object; home study SharePoint ToolShell	Confirmed	[72][2]
27 Aug 2025	AA25-239A: 23 agencies in 13 countries on Salt Typhoon and three named companies	G1045	Confirmed	[53]
Sep 2025	AI-orchestrated campaign attributed to GTG-1002	C0062	Confirmed	[2]
4 Dec 2025	CISA, NSA and the Canadian Cyber Centre attribute BRICKSTORM to PRC state-sponsored actors	No ATT&CK group object; BRICKSTORM is S9015	Confirmed	[73]
Dec 2025	UK sanctions Integrity Technology Group and one other company	Flax Typhoon, no ATT&CK object	Confirmed	[74]
25 Feb 2026	GTIG disrupts UNC2814 and its GRIDTIDE backdoor	No ATT&CK object	Confirmed	[75]
23 Apr 2026	AA26-113A and NCSC at CYBERUK on covert networks of compromised devices	G1017; Flax Typhoon	Confirmed	[76][77][74]
25 Apr 2026	Xu Zewei extradited to the US and held in Houston, per his lawyer	G0125	Reported	[92]
15 Jun 2026	GTIG publishes the UNC6508 REDCap campaign	No ATT&CK object	Confirmed	[78]
late Aug 2026	ARMORY CASTLE (formerly UNC5221) exploits a BeyondTrust Remote Support zero-day (GTIG, 15 Oct 2026)	No ATT&CK object	Confirmed	[82]
26 Aug 2026	DOJ seizure affidavit against QTFY actors, described in the affidavit as Chinese hackers; press report three domains seized and the platforms run by Nanjing Xinjiuwei	No ATT&CK binding	Confirmed (affidavit); seizure count and company Reported	[94][95]
8 Sep 2026	AA26-251A on industrial-scale model distillation by China-based AI companies	Not an intrusion campaign	Confirmed	[79]
8 Oct 2026	AA26-281A and NCSC on actors enabled by Integrity Tech, including US Healthcare and Public Health victims	Flax Typhoon, no ATT&CK object	Confirmed	[80][81]

Part 4: the actors behind the headlines

APT1 and the PLA (G0006)

APT1 is the case that set the template. Mandiant's report of 19 February 2013 linked the group to the PLA General Staff Department's 3rd Department, Military Cover Designator 61398, documented economic espionage since 2006 against 141 victims and released over 3,000 indicators (Confirmed [55]). Fifteen months later a W.D. Pa. grand jury indicted five officers of Unit 61398, Wang Dong, Sun Kailiang, Wen Xinyu, Huang Zhenyu and Gu Chunhui, on 31 counts covering six US victims in nuclear power, metals and solar: Westinghouse, SolarWorld, US Steel, ATI, the United Steelworkers and Alcoa (Confirmed [18]). ATT&CK records 23 techniques and 17 software for the group (Confirmed [3]). The Equifax case six years later followed the same pattern with a different PLA body: four members of the 54th Research Institute were charged with exploiting Apache Struts on Equifax's dispute portal, taking PII on about 145 million Americans, routing through about 34 servers in nearly 20 countries and wiping logs every day (Confirmed [24]).

APT10, menuPass and Cloud Hopper (G0045)

menuPass has been active since at least 2006, and ATT&CK says its members acted with the MSS Tianjin State Security Bureau and worked for Huaying Haitai; its targets include healthcare and biotechnology with an emphasis on Japan, and managed service providers in 2016 and 2017 (Confirmed [16]). BAE Systems named the MSP campaign Operation Cloud Hopper on 4 April 2017 (Confirmed [58]). On 20 December 2018 DOJ charged Zhu Hua and Zhang Shilong with hacking MSPs and more than 45 US technology companies from about 2006 to 2018; the FBI notice adds MSPs with clients in at least 12 countries and PII of over 100,000 Navy personnel (Confirmed [21][90]). The same day the UK and its allies said APT10 acted for the MSS: NCSC assessed "with the highest level of probability" that APT10 ran the campaign (Confirmed [22]). MirrorFace (G1054) is now placed under the menuPass umbrella in ATT&CK, and Japan's National Police Agency and NISC described its campaign since about 2019 against Japanese organizations as suspected of Chinese involvement (this study's translation) and aimed at national security and advanced technology information (Confirmed [1][45]).

APT40, Leviathan and Hainan Xiandun (G0065)

Leviathan is the best-documented MSS bureau operation in the record (Inferred). ATT&CK attributes it to the Hainan State Security Department and an affiliated front company, active since at least 2009 (Confirmed [14]). The S.D. Cal. indictment, returned in May 2021 and unsealed in July, names HSSD officers Ding Xiaoyang, Cheng Qingmin and Zhu Yunmin and hacker Wu Shurong, operating through Hainan Xiandun from 2011 to 2018 (Confirmed [28][87]). AA21-200A, published the same day, gives the TTPs: T1190, T1133, T1566.001 and T1566.002, T1189, T1078, T1505.003, T1090.003, T1572, T1560 and T1041 (Confirmed [29]). Three years later ASD's ACSC led an eight-country advisory, AA24-190A, attributing APT40 to the MSS and documenting rapid weaponization of public proof-of-concept exploits, web shells and SOHO relays; it cites the Hainan basing only as prior reporting, a weaker statement than AA21-200A (Confirmed [37]). ATT&CK records the 2022 Australian case studies as campaign C0049 (Confirmed [14]).

APT31, ZIRCONIUM and the Hubei bureau (G0128)

On 25 March 2024 the United States and the United Kingdom acted together. E.D.N.Y. charged seven APT31-linked PRC nationals, Ni Gaobin, Weng Ming, Cheng Feng, Peng Yaowen, Sun Xiaohui, Xiong Wang and Zhao Guangzong, over about 14 years of targeting critics, businesses and officials with more than 10,000 malicious emails (Confirmed [34]). OFAC sanctioned Wuhan Xiaoruizhi, a front company the MSS Hubei State Security Department set up in 2010, with Zhao and Ni; named victims include defense industrial base firms, a Texas energy company in 2018 and a California MSP, and targets included White House, Justice, Commerce, Treasury and State staff, Congress and naval schools (Confirmed [35]). NCSC assessed it almost certain that APT31 ran reconnaissance against UK parliamentarians in 2021, and that an unnamed China state-affiliated actor highly likely accessed Electoral Commission email and Electoral Register data in 2021 and 2022 (Confirmed [65][36]). On 28 May 2025 the Czech government named the PRC as responsible for a campaign since 2022 against an unclassified Foreign Ministry network, an institution designated as critical infrastructure, carried out by APT31 (Confirmed [50]); NUKIB says the attribution rests on joint analysis with Military Intelligence, the Office for Foreign Relations and Information and the Security Information Service (Confirmed [51]). ATT&CK also links ZIRCONIUM to 2020 US election targets and to campaign C0053 FLORAHOX Activity (Confirmed [1][2]).

APT27, i-Soon and the data brokers (G0027, G0143)

Threat Group-3390 (APT27, Emissary Panda, LuckyMouse, Linen Typhoon) is known for strategic web compromises since at least 2010 against aerospace, government, defense, technology, energy, manufacturing and gambling (Confirmed [13]). The March 2025 charges reframed it as part of a market. DOJ charged eight i-Soon employees, two Ministry of Public Security officers, and APT27 actors Yin Kecheng and Zhou Shuai; i-Soon allegedly sold data to at least 43 MSS or MPS bureaus in 31 provinces, and Yin and Zhou allegedly targeted victims including health care systems from 2013 to 2024 using PlugX (Confirmed [47]). The FBI's notice on i-Soon (tracked as Aquatic Panda) lists victims from about 2016 to 2023: a large US religious organization, PRC critics and dissidents, a state legislative body, US agencies, Asian foreign ministries and news organizations, with no healthcare victims named (Confirmed [96]). Treasury says Zhou has brokered stolen data since at least 2018, with victims including an academic health system affiliated with a university (Confirmed [49]).

HAFNIUM, Silk Typhoon and the Exchange campaign (G0125)

Microsoft attributed exploitation of four Exchange Server zero-days (CVE-2021-26855, CVE-2021-26857, CVE-2021-26858 and CVE-2021-27065) with high confidence to HAFNIUM on 2 March 2021, naming infectious disease researchers, law firms, higher education, defense contractors, think tanks and NGOs as targets (Confirmed [59]). On 19 July 2021 the US, joined by the EU, UK and NATO, attributed the campaign with a high degree of confidence to MSS-affiliated actors (Confirmed [60]); NCSC called it "almost certain" a Chinese state-backed actor was behind it (Confirmed [30]). The 2025 arrest of Xu Zewei in Milan connects the campaign to a named contractor: DOJ says the Shanghai SSB directed Xu, who worked at Shanghai Powerock, from February 2020 to June 2021, and the FBI says over 60,000 US entities were targeted and more than 12,700 victimized (Confirmed [52]). In March 2025 Microsoft described Silk Typhoon, its current name for HAFNIUM, abusing stolen API keys and credentials of IT providers since late 2024 to reach downstream tenants, with healthcare among the targets (Confirmed [69]).

The 2021 Exchange mass exploitation is covered in its own study, Operation Exchange Marauder; this section summarizes the actor record only.

APT41 (G0096): summary of the home study

APT41 (Wicked Panda, Brass Typhoon, BARIUM) is assessed by researchers, per ATT&CK, as a Chinese state-sponsored espionage group that also runs financially motivated operations, active since at least 2012 and targeting industries including healthcare in 14 countries, with 82 techniques, 32 software and campaigns C0017 (May 2021 to February 2022) and C0040 APT41 DUST (January 2023 to June 2024) (Confirmed [9]). DOJ charged five PRC nationals and two Malaysian businessmen in September 2020 over more than 100 victims (Confirmed [27]), and HC3 lists its health-sector attacks from 2014 to 2022 (Confirmed [86]). The full record, tooling and timeline are in APT41: The Public Record.

Volt Typhoon and pre-positioning (G1017)

Microsoft introduced Volt Typhoon on 24 May 2023 as a China-based state-sponsored actor active since mid-2021, targeting US critical infrastructure including Guam, and assessed with moderate confidence that it was developing capabilities to disrupt US-Asia communications in a crisis (Confirmed [62]). Initial access came through internet-facing Fortinet FortiGuard devices, and traffic was proxied through compromised ASUS, Cisco, D-Link, NETGEAR and Zyxel SOHO devices (Confirmed [62]). AA23-144A, from NSA, CISA, FBI and the Five Eyes, described the same actor living off the land (Confirmed [31]). In December 2023 a court-authorized operation removed KV Botnet malware from hundreds of US SOHO routers, mostly end-of-life Cisco and NetGear devices, and FBI Director Wray described the activity as pre-positioning to cause harm in a conflict (Confirmed [32]). AA24-038A, written by CISA, NSA, FBI and their US co-authors and co-released with partners in Australia, Canada, the UK and New Zealand, then gave the US authoring agencies' assessment, with high confidence, that Volt Typhoon is pre-positioning on IT networks to reach and disrupt OT in Communications, Energy, Transportation and Water, with some footholds held at least five years; it names CVE-2022-42475 on a FortiGate 300D among the entry points (Confirmed [33]). ATT&CK records campaigns C0035 KV Botnet Activity and C0039 Versa Director Zero Day Exploitation and software including S1144 FRP and S1154 VersaMem (Confirmed [10]).

Salt Typhoon and the telecoms (G1045)

The Congressional Research Service records that media first reported in early October 2024 that PRC state-sponsored hackers had infiltrated US telecoms and ISPs (Confirmed [67]). On 13 November 2024 the FBI and CISA confirmed PRC-affiliated actors had compromised multiple telecom companies, stealing customer call records, compromising private communications of a limited number of people in government or politics, and copying information subject to court-ordered law enforcement requests (Confirmed [40]). The December 2024 hardening guide said the exploitation matched existing weaknesses and no novel activity was observed (Confirmed [41]). Treasury sanctioned Sichuan Juxinhe in January 2025 for direct involvement in Salt Typhoon's exploitation of US telecoms (Confirmed [46]). AA25-239A, co-sealed by 23 agencies in 13 countries, names Sichuan Juxinhe, Beijing Huanyu Tianqiong and Sichuan Zhixin Ruijie, links them to PLA units and the MSS, and lists the initial-access CVEs: CVE-2024-21887 (with CVE-2023-46805), CVE-2024-3400, CVE-2023-20198, CVE-2023-20273 and CVE-2018-0171; zero-day use was not observed (Confirmed [53]). The live ATT&CK page lists one software entry, S1206 JumbledPath, and 58 Enterprise techniques matching the library exactly (Confirmed [12]).

Flax Typhoon and Integrity Technology Group (no ATT&CK object)

Flax Typhoon is the clearest case of a Chinese company running infrastructure as a service (Inferred). A court-authorized operation in September 2024 disrupted a botnet of more than 200,000 consumer devices built and run by Integrity Technology Group, a publicly traded Beijing company, which sold access through a web application labelled KRLab; someone attempted a DDoS against the FBI's remediation infrastructure (Confirmed [38]). The FBI, CNMF and NSA advisory put the botnet at over 260,000 devices as of June 2024, active since mid-2021, with activity consistent with Flax Typhoon, RedJuliett and Ethereal Panda (Confirmed [39]). OFAC designated the company in January 2025 (Confirmed [44]), the UK sanctioned it in December 2025 (Confirmed [74]), and AA26-281A in October 2026 described actors enabled by Integrity Tech combining automated and hands-on tools against US Healthcare and Public Health, government, critical manufacturing and IT, plus healthcare systems in Southeast Asia (Confirmed [80]). ATT&CK has no group record for Flax Typhoon in the library data (Confirmed [39][6]).

UNC5221, ARMORY CASTLE and the edge zero-days (no ATT&CK group object; BRICKSTORM is S9015)

Mandiant attributes exploitation of the Ivanti Connect Secure zero-day CVE-2025-0282, beginning mid-December 2024, to UNC5221, a suspected China-nexus espionage actor (Confirmed [68]). GTIG attributes BRICKSTORM intrusions to UNC5221 and related clusters, with a 393-day average dwell, and states it does not consider UNC5221 the same as Silk Typhoon (Confirmed [70]). CISA, NSA and the Canadian Cyber Centre went further in December 2025, attributing BRICKSTORM on VMware vCenter and Windows to PRC state-sponsored actors (Confirmed [73]). In October 2026 GTIG attributed late-August 2026 zero-day exploitation of BeyondTrust Remote Support, listed as CVE-2026-102764, to ARMORY CASTLE, its new name for UNC5221 (Confirmed [82]). ATT&CK records the 2023 to 2024 Ivanti activity as campaign C0029 Cutting Edge (Confirmed [2]).

UNC3886 and the routers (G1048)

UNC3886 is a China-nexus espionage group active since at least 2022 that exploits edge and virtualization zero-days against defense, technology and telecoms in the US and Asia-Pacific (Confirmed [15]). Mandiant found TINYSHELL-based backdoors on Juniper MX routers in mid-2024, involving CVE-2025-21590 (Confirmed [71]), and ATT&CK records the activity as campaign C0056 RedPenguin (July 2024 to March 2025) (Confirmed [15]).

The others

- **Mustang Panda (G0129)** has the largest footprint in ATT&CK: 85 techniques and 23 software, 15 aliases including RedDelta, BRONZE PRESIDENT, EARTH PRETA, TWILL TYPHOON and UNC6384, and campaign C0047 (July 2023 to December 2024) (Confirmed [8]).
- **Chimera (G0114)** is a suspected China-based group targeting Taiwan's semiconductor industry and airline data since at least 2018; 59 techniques (Confirmed [11]).
- **Ke3chang (G0004)**, also APT15, NICKEL and Nylon Typhoon, targets oil, government, diplomatic, military and NGO organizations in the Americas and Europe since at least 2010 (Confirmed [17]); Microsoft's DCU seized its websites in December 2021, citing targets in the US and 28 other countries (Confirmed [61]).
- **APT3 (G0022)** is attributed by researchers to the MSS (Confirmed [1]); the Boyusec indictment names the company but not the APT3 label (Confirmed [19]).

- **Storm-0558** (now Antique Typhoon) is assessed with moderate confidence by Microsoft to be China-based; it accessed mail at about 25 organizations with forged tokens from May 2023 (Confirmed [64]). No ATT&CK object.
- **UNC2814**, a suspected PRC-nexus group tracked since 2017, used the GRIDTIDE backdoor with Google Sheets API command and control; GTIG counted 53 confirmed victims in 42 countries as of 18 February 2026, mainly telecoms and government, with no observed overlap with Salt Typhoon (Confirmed [75]).
- **Sichuan Silence** and Guan Tianfeng used the Sophos zero-day against about 81,000 firewalls; over 23,000 were in the US, 36 of them protecting critical infrastructure, and the malware was modified to deploy ransomware (Ragnarok) if victims remediated (Confirmed [42][43]).

Part 5: healthcare and life sciences

What the record shows

China-attributed actors have targeted healthcare, pharmaceuticals, medical devices and biomedical research for more than a decade, mostly for intellectual property and identity data rather than disruption. Nine groups in the ATT&CK data carry healthcare or life-sciences targeting in their descriptions: menuPass (healthcare, biotechnology), Leviathan (biomedical, healthcare), APT41 (healthcare), Tonto Team (healthcare), Deep Panda (Anthem), HAFNIUM (infectious disease researchers), APT19 (pharmaceutical), Earth Lusca (COVID-19 research) and BRONZE BUTLER (biotechnology), and campaign C0058 ToolShell lists healthcare among its sectors (Confirmed [1]).

This study opened three HHS Health Sector Cybersecurity Coordination Center (HC3) briefings on China-based threats to the Healthcare and Public Health (HPH) sector:

Date	HC3 product	What it says	Source
19 Nov 2020	Threat briefing, Chinese state-sponsored cyber activity	At least 8 China-attributed APT groups have targeted HPH: APT41, APT24 (PittyTiger), APT22, APT20 (Twivy), APT18 (Wekby), APT10 (menuPass), APT9 (Nightshade Panda), APT1 (Unit 61398). APT22 targeted healthcare, biomedical and pharmaceutical organizations, including multi-year targeting of a cancer-research health center; APT9 was historically very active against pharmaceuticals and biotechnology; APT20 targeted healthcare and MSPs and bypassed VPN two-factor authentication; APT24 targeted healthcare in the US and Taiwan	[83]
6 May 2021	China's Five-Year Plans and the HPH	Chinese APTs that targeted HPH entities: APT1, APT10, APT18, APT41, APT24, Deep Panda, APT22, APT20, APT9. Ties targeting to the Five-Year Plans; APT10 targeted a Japanese pharmaceutical company in 2019 to 2020	[85]
16 Aug 2023	Threat Profile, China-based threat actors	APT41, APT10, APT18 and APT22 are the greatest China-based threats to US HPH, with APT27 in the targeting matrix; APT18 targets biotech, pharmaceutical, cancer research and healthcare manufacturing since 2013 and is believed affiliated with the PLA Navy; APT10 used healthcare-themed phishing documents in Japan	[86]

The ATT&CK binding for the HC3 names: G0026 APT18, G0045 menuPass, G0096 APT41, G0006 APT1, G0009 Deep Panda and G0027 Threat Group-3390 have pages; APT22, APT20 and APT9 have no ATT&CK object (Confirmed [6]). HC3 equates APT22 with Suckfly and G0039, but ATT&CK lists no APT22 alias for

G0039 Suckfly, so this study does not bind APT22 to G0039 (Inferred [86][1]). HC3 also equates APT24 with PittyTiger, but ATT&CK lists no APT24 alias for G0011, so this study does not bind APT24 to G0011 either (Inferred [83][1]). APT18 is in the library as G0026 but its ATT&CK description carries no China wording, which is why it sits outside the Part 1 counts (Confirmed [1]).

The healthcare timeline

Date	Event	Actor	Grade	Source
2006 to 2018	APT10 indictment lists target industries including healthcare, medical equipment, biotechnology and pharmaceutical manufacturing (the FBI wanted notice does not list healthcare)	G0045	Confirmed	[21][90]
2011 to 2018	APT40 steals Ebola virus and vaccine research, including genetic-sequencing technology; victims include a virus treatment and vaccine facility in California and Florida, a university school of medicine and NIH	G0065	Confirmed	[87][88]
Sep 2012 to Jul 2015	APT40 overt acts: malware on a school-of-medicine system (Sep 2012) and on Research Facility A's Ebola program (Nov 2012, 25 Jul 2014); in 2015 the conspirators sought information on tularemia, Marburg vaccine, Ebola, US national health security and HIV/AIDS strategies	G0065	Confirmed	[87]
Feb 2014 to Jan 2015	Anthem: about 78.8 million people's records, including health identification numbers and Social Security numbers; DOJ and the FBI allege a China-based hacking group but no state direction	G0009 per ATT&CK	Confirmed	[23][89][1]
Apr and Jun 2014	Community Health Systems: names, addresses, birthdates, phone numbers and SSNs of about 4.5 million people, no medical, clinical or card data; attributed to an APT group originating from China	No group in the filing; APT18 per HC3	Confirmed (filing); APT18 Reported	[56][83]
Jul 2014 to 2022	APT41 health-sector attacks: a medical devices subsidiary (2014 to 2016), a biotech being acquired (2015), a cancer-research center (2018), Citrix, Zoho and Cisco exploitation in health (2021), at least USD 20 million in COVID relief funds (2022)	G0096; home study	Confirmed (HC3)	[86]
Apr 2019	Suspected Chinese espionage actors target a US cancer-research health center	Unnamed; APT41 and APT22 earlier	Confirmed (HC3)	[83]
Jan to Jun 2020	Li Xiaoyu and Dong Jiazhi: a Maryland biotech announcing COVID-19 vaccine work (Jan), a California biotech on COVID-19 antivirals (Feb), a California diagnostics firm making COVID-19 test kits (May), a Massachusetts cancer-treatment biotech (Jun)	No ATT&CK binding	Confirmed	[26][83]
Feb 2020	Xu Zewei targets COVID-19 research mailboxes at a Texas university; press add immunologists and virologists	G0125	Confirmed (DOJ); detail Reported	[52][91]
13 May 2020	FBI and CISA: the FBI is investigating PRC targeting of US COVID-19 research for vaccine, treatment and testing data	PRC actors	Confirmed	[25][84]
13 Aug 2020	Medical Center 1 of Ohio reports an actor-leased server targeting its Pulse Secure VPN (CVE-2019-11510); a Missouri insurance agency was targeted via CVE-2019-19781 in Jan 2020	QTFY actors, no ATT&CK binding	Confirmed	[94]
2 Mar 2021	HAFNIUM's targets include infectious disease researchers	G0125	Confirmed	[59][1]
19 Jul 2021	AA21-200A lists healthcare and biomedical among APT40's victims; AA21-200B names medical institutions among PRC targets	G0065	Confirmed	[29][93]
2013 to 2024	Yin Kecheng and Zhou Shuai target victims including health care systems; Zhou's victims include an academic health system	G0027	Confirmed	[47][49]

Date	Event	Actor	Grade	Source
Sep 2023 to Nov 2025	UNC6508 compromises a medical research institution's REDCap server, uses INFINITERED to capture credentials and abuses content compliance rules to exfiltrate mail; targets include clinical providers, academic medical centers, military health and health regulators in North America; GTIG grades the attribution high confidence	No ATT&CK object	Confirmed	[78]
Sep 2024	QTFY actors intrude at NIH and an HHS agency via an Ivanti Cloud Services Appliance zero-day	No ATT&CK binding	Confirmed	[94]
late 2024 onward	Silk Typhoon targets healthcare among other sectors through IT providers, RMM companies and MSPs	G0125	Confirmed	[69]
Jul 2025	SharePoint ToolShell exploitation, healthcare among the sectors	C0058	Confirmed	[2]
23 Apr 2026	AA26-113A highest-threat guidance on covert networks names healthcare with energy, transport, digital infrastructure and government	Covert networks	Confirmed	[76]
26 Aug 2026	Press account of DOJ: targets include US military networks, hospitals, power companies and defense contractors, 2018 to 2026	QTFY actors	Reported	[95]
8 Oct 2026	AA26-281A: US Healthcare and Public Health victims, plus healthcare systems in Southeast Asia, with access since mid-January 2021	Integrity Tech, Flax Typhoon	Confirmed	[80]

What the record does not show

Volt Typhoon, the actor most associated with disruption, does not list healthcare among its primary sectors: AA24-038A names Communications, Energy, Transportation Systems and Water and Wastewater (Confirmed [33]). Salt Typhoon targets telecommunications, government, transportation, lodging and military networks (Confirmed [53]). No source in this study attributes a disruptive or ransomware attack on a hospital to a Chinese state actor; the healthcare record is theft of research, identity data and mail (Inferred from [1][23][56][78][86]). The 2024 Change Healthcare ransomware attack, a criminal operation, is covered in its own study, Change Healthcare 2024. Sichuan Silence's plan to deploy ransomware if victims remediated, and Cinnamon Tempest's Babuk-derived ransomware, show the capability exists in the ecosystem (Confirmed [42][1]).

The healthcare defender's watch list

Ranked by how recent and how health-specific the primary record is (Inferred ranking from the cited sources):

1. **UNC6508** for medical research institutions, academic medical centers, military health and regulators: REDCap and other research applications, credential capture, mail rule abuse [78].
2. **Actors enabled by Integrity Technology Group (Flax Typhoon and related)** for US HPH organizations: automated scanning and covert device networks [80][76].
3. **Silk Typhoon (HAFNIUM)** for any hospital that relies on an IT provider, MSP or RMM tool: stolen API keys and provider credentials [69].
4. **APT41** for medical devices, biotech and pharmaceuticals, per HC3 and its home study [86].
5. **APT40** for biomedical and infectious disease research [87][37].

6. **APT27 and the brokers (Yin, Zhou, i-Soon)** for health systems, especially university-affiliated ones [47][49].

Part 6: tooling and exploited vulnerabilities

Shared software

ATT&CK's software records show a shared toolkit across the China-attributed groups, dominated by built-in Windows tools and a few long-lived malware families. The table lists every software object ATT&CK v20.0 records for six or more of the 46 groups (Confirmed [3]):

Software	Type	Groups (of 46)	Notes
S0039 Net	Tool	15	Built-in Windows tool
S0002 Mimikatz	Tool	14	Credential dumping
S0013 PlugX	Malware	12	Used by Yin and Zhou per DOJ [47]
S0154 Cobalt Strike	Malware	12	Named in AA21-200B [93]
S0012 PoisonIvy	Malware	11	
S0057 Tasklist	Tool	9	Built-in
S0029 PsExec	Tool	8	
S0097 Ping	Tool	8	Built-in
S0100 ipconfig	Tool	8	Built-in
S0032 gh0st RAT	Malware	7	
S0357 Impacket	Tool	7	
S0596 ShadowPad	Malware	7	
S0020 China Chopper	Malware	6	Web shell
S0104 netstat	Tool	6	Built-in

Volt Typhoon's ATT&CK record adds S0108 netsh, S0359 Nltest and S0645 Wevtutil, all built in, plus FRP and VersaMem (Confirmed [10]). Salt Typhoon's single software entry is JumbledPath (Confirmed [12]). BRICKSTORM is in ATT&CK as S9015, with no group attached (Confirmed [3]). The vendor reports name further families with no ATT&CK object in the library data: GRIDTIDE, INFINITERED, SAWFLY and BARECLAW, and the TINYSHELL-based Juniper backdoors (Confirmed [70][75][78][82][71]).

Exploited vulnerabilities

Five joint advisories name 43 distinct initial-access CVEs between them, and every one is in CISA's KEV catalog (version 2026.11.03, 1,767 entries); 29 are marked known for ransomware campaign use, 21 were added on 3 November 2021, the catalog's first day, and the most recent additions are CVE-2024-3400 (12 April 2024) and CVE-2024-21887 with CVE-2023-46805 (10 January 2024) (Confirmed [100]).

Advisory	CVEs it names	Source
AA22-158A (7 Jun 2022), network devices	Cisco CVE-2018-0171, CVE-2019-15271, CVE-2019-1652; Citrix CVE-2019-19781; DrayTek CVE-2020-8515; D-Link CVE-2019-16920; Fortinet CVE-2018-13382; MikroTik CVE-2018-14847; Netgear CVE-2017-6862; Pulse CVE-2019-11510, CVE-2021-22893; QNAP CVE-2019-7192 to CVE-2019-7195; Zyxel CVE-2020-29583	[97]
AA22-279A (6 Oct 2022), top 20	Log4j CVE-2021-44228; Pulse CVE-2019-11510; GitLab CVE-2021-22205; Atlassian CVE-2022-26134 and CVE-2021-26084; Exchange CVE-2021-26855, CVE-2021-26857, CVE-2021-26858, CVE-2021-27065; F5 CVE-2020-5902 and CVE-2022-1388; vCenter CVE-2021-22005; Citrix CVE-2019-19781; Cisco HyperFlex CVE-2021-1497; Buffalo CVE-2021-20090; Hikvision CVE-2021-36260; Sitecore CVE-2021-42237; APISIX CVE-2022-24112; Zoho CVE-2021-40539; Apache HTTP CVE-2021-41773	[99]
AA24-038A (7 Feb 2024), Volt Typhoon	CVE-2022-42475 on a FortiGate 300D; Fortinet, Ivanti Connect Secure, NETGEAR, Citrix and Cisco products	[33]
AA24-190A (8 Jul 2024), APT40	CVE-2021-44228, CVE-2021-31207, CVE-2021-26084, CVE-2021-34523, CVE-2021-34473	[37]
AA25-239A (27 Aug 2025), Salt Typhoon	CVE-2024-21887 (with CVE-2023-46805), CVE-2024-3400, CVE-2023-20198, CVE-2023-20273, CVE-2018-0171	[53]

Vendor reports add zero-days: the four Exchange CVEs (HAFNIUM, 2021) [59]; CVE-2025-0282 in Ivanti Connect Secure (UNC5221, December 2024) [68]; CVE-2025-21590 in Juniper Junos (UNC3886) [71]; the SharePoint chain CVE-2025-49704, CVE-2025-49706, CVE-2025-53770 and CVE-2025-53771 (July 2025) [72]; and CVE-2026-102764 in BeyondTrust Remote Support (ARMORY CASTLE, August 2026) [82]. Microsoft lists CVE-2023-3519, CVE-2024-3400 and CVE-2025-0282 for Silk Typhoon (Confirmed [69]). The Log4Shell CVE is covered in its own study, Log4Shell (CVE-2021-44228). The FBI, CNMF and NSA advisory on the Integrity Tech botnet lists 67 CVE entries in an appendix, many on devices that are likely still vendor-supported (Confirmed [39]).

Part 7: the government record

Attribution statements by government

Date	Government	Statement	Confidence words	Source
20 Dec 2018	UK and allies	APT10 acted for the MSS; ran Cloud Hopper against MSPs since at least 2016	"highest level of probability" for APT10; MSS relationship highly likely	[22]
19 Jul 2021	US with the EU, UK and NATO	Exchange zero-day campaign by MSS-affiliated actors; MSS-linked operators ran ransomware for profit	"high degree of confidence"	[60]
19 Jul 2021	UK	Exchange compromise by a Chinese state-backed actor; APT40 sponsored by the Hainan SSD; APT31 contractors for the MSS	"almost certain"; "highly likely"; "likely" (the press text is flat, the NCSC table is hedged)	[30]
25 Mar 2024	UK	APT31 reconnaissance of parliamentarians; Electoral Commission compromise by a China state-affiliated entity	"almost certain"; "highly likely"	[36][65]
8 Jul 2024	Australia with the US, UK, Canada, New Zealand, Germany, South Korea, Japan	APT40 conducts operations for the MSS	Assessment	[37]
8 Jan 2025	Japan (NPA, NISC)	MirrorFace campaign since about 2019	suspected of Chinese involvement (this study's translation of the Japanese text)	[45]
28 May 2025	Czech Republic	PRC responsible for the Foreign Ministry campaign; APT31 carried it out	National attribution; NUKIB: "most likely" through APT31	[50][51]
27 Aug 2025	23 agencies in 13 countries	PRC state-sponsored actors; three companies supply PLA and MSS	Joint assessment	[53]
4 Dec 2025	US and Canada	BRICKSTORM by PRC state-sponsored actors	Attribution	[73]
23 Apr 2026	UK with 15 partners	Covert networks used by the majority of China-linked actors	"believed"	[74]
8 Oct 2026	US, UK, Australia, Canada, Japan, New Zealand, Spain	Actors enabled by Integrity Tech; activity "consistent with" Flax Typhoon, Ethereal Panda and Red Juliett	"consistent with"	[80][81]

Treasury sanctions

Date	Designation	Tie	Source
25 Mar 2024	Wuhan Xiaoruizhi, Zhao Guangzong, Ni Gaobin	APT31, MSS Hubei SSD front	[35]
10 Dec 2024	Sichuan Silence, Guan Tianfeng	Sophos firewall campaign; PRC intelligence clients	[43]
3 Jan 2025	Integrity Technology Group	Flax Typhoon, summer 2022 to fall 2023	[44]
17 Jan 2025	Yin Kecheng, Sichuan Juxinhe	Treasury compromise; Salt Typhoon	[46]
5 Mar 2025	Zhou Shuai, Shanghai Heiying	Data brokering since 2018	[49]

Disruptions

Three court-authorized or court-ordered actions in the record took infrastructure away from China-attributed actors: Microsoft's seizure of NICKEL websites (December 2021) [61], the KV Botnet removal (December 2023) [32], and the Integrity Tech botnet disruption (September 2024) [38]. A fourth is the August 2026 DOJ seizure affidavit against QTFY actors, described in the affidavit as Chinese hackers (Confirmed [94]); press report three domains seized and the platforms run by Nanjing Xinjiuwei (Reported [95]). GTIG's disruption of UNC2814 in February 2026 was a private-sector action (Confirmed [75]).

China's response

China's National Computer Virus Emergency Response Center and a national laboratory issued a third report in October 2024, after reports in April and July, calling Volt Typhoon a "political farce" and a US false flag operation. This study saw it only through PRC state media, not the report itself (Reported [63]).

Part 8: the defender view

What the advisories ask defenders to watch

Read together, the advisories converge on six detection threads (Inferred synthesis of [29][31][33][37][41][53][97]):

1. **Edge and network-device configuration drift and integrity**: compare running configurations to baselines; check ACLs, local accounts and routing tables; look for guest containers on network operating systems, management services on non-standard ports, TACACS+ to unapproved servers and unexpected GRE or IPsec tunnels; verify firmware and image hashes; capture live artifacts before eradication (Confirmed [53]). Techniques: T1584.008, T1602, T1599.
2. **NTDS and LSASS credential access**: ESENT application log events for NTDS.dit copying; LSASS protection and Credential Guard; attack surface reduction rules against LSASS theft and PsExec or WMI process creation (Confirmed [31][33][62]). Techniques: T1003.001, T1003.003.
3. **Living-off-the-land command lines**: process creation auditing with command line (event 4688), deep PowerShell logging and WMI tracing, against behavioral baselines (Confirmed [31][33]). The joint LOTL guidance warns that many organizations cannot yet tell LOTL activity from legitimate behavior, and AA23-144A warns not to assume findings are malicious without investigation (Confirmed [98][31]). Techniques: T1059.001, T1059.003, T1047.

4. **VPN and RDP logon anomalies**: impossible travel on VPN logons, RDP session review, logon success and failure review for password spraying (Confirmed [31][33]). Techniques: T1078, T1133, T1021.001, T1110.003.
5. **Web shells and execution from world-writable folders**: AA24-190A gives three Sigma rules for execution from world-writable locations, and AA24-038A asks for checks of world-writable folders for unexpected files (Confirmed [37][33]). Technique: T1505.003.
6. **Log clearing**: investigate every Security log clear (event 1102) (Confirmed [33]). Technique: T1685.005.

What the advisories ask defenders to harden

- **Patch the edge first.** AA24-038A: prioritize known exploited vulnerabilities in internet-facing systems and the targeted appliances (Confirmed [33]). AA24-190A: patches or mitigations for internet-facing devices within 48 hours (Confirmed [37]). AA22-279A: patch KEV-first (Confirmed [99]).
- **Replace end-of-life devices.** Named in AA24-038A, AA24-190A, AA25-239A, AA22-158A and AA22-279A; most KV Botnet routers were end-of-life Cisco and NetGear devices (Confirmed [33][37][53][97][99][32]).
- **Isolate management.** An isolated management VRF or out-of-band network with default-deny ACLs; SSHv2 and SNMPv3 only; public-key authentication; disable Smart Install; block outbound traffic from management interfaces (Confirmed [53][97]). Keep SOHO and edge management interfaces off the internet (Confirmed [31][62]).
- **Phishing-resistant MFA and tiered administration.** No credentials stored on edge devices; reset krbtgt twice on suspected compromise; default-deny IT and OT segmentation (Confirmed [33]).
- **Centralized, encrypted logging**, including AAA command accounting on network devices and web, Windows event and proxy logs (Confirmed [53][37]).
- **For consumer and SOHO devices**: disable UPnP, remote management and file sharing; segment IoT; monitor traffic volume; patch firmware; replace default passwords (Confirmed [39]).

The vulnerability-management view

For a VM program, the China advisories reduce to an edge-first rule: the KEV due-date queue for internet-facing VPN, firewall, router and web-application CVEs is the control most aligned with the record, and AA24-190A's 48-hour window is the government benchmark for time-to-mitigate (Inferred [37][100]). All 43 initial-access CVEs named across the five advisories are in KEV as of catalog version 2026.11.03 (Confirmed [100]), which means a program that closes KEV edge findings quickly covers the documented entry points; zero-day entries such as CVE-2025-0282 and the BeyondTrust flaw are the residual risk, and the advisories' answer to them is monitoring and isolation of the device rather than patch speed (Inferred [68][82][53]). Phishing still matters: Spearphishing Attachment is the sixth most shared technique across the 46 groups in ATT&CK's records (Confirmed [3]).

For hospitals and research institutions

Apply the edge rule first, then the health-specific record (Inferred from Part 5):

- Inventory research applications such as REDCap, put them behind the same edge patching and authentication rules as clinical systems, and audit mail transport and compliance rules for changes that

copy or forward mail (Confirmed [78] for the abuse; the control is Inferred).

- Review every IT service provider, MSP and RMM integration, rotate API keys and provider credentials, and log provider access to tenants (Confirmed [69] for the access path; the control is Inferred).
- Include academic medical centers' university networks in scope: the APT40 and Zhou Shuai records name university schools of medicine and academic health systems (Confirmed [87][49]).

Part 9: MITRE mappings

This Part binds the verified record to MITRE ATT&CK v20.0, CAPEC, CWE and D3FEND, with every id checked against the live MITRE page or the library's data files on 3 November 2026. None of the 47 China-attributed groups appears in the library's ATT&CK for ICS or Mobile group files, so there is no ICS or Mobile table (Confirmed [7]). Volt Typhoon's OT intent is an assessment about future disruption, not a recorded ICS technique (Confirmed [33]).

ATT&CK Enterprise techniques named in government advisories

Each id below is named in at least one joint government advisory on PRC actors and checked against `data/attack/technique_profiles.jsonl` (Confirmed [101]). The advisories name more techniques than this table; it lists the subset this study verified. Ids that the source reported under an earlier ATT&CK version and that are revoked in v20.0 are shown under their successor from `data/attack/superseded_by.jsonl`.

Id	Name	Tactic (v20.0)	Where it appears in the record	Source
T1595.002	Vulnerability Scanning	Reconnaissance	AA22-158A (PRC actors, network devices, 7 Jun 2022)	[97]
T1584.005	Botnet	Resource Development	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1584.008	Network Devices	Resource Development	AA24-190A (APT40, 8 Jul 2024); AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[37][53]
T1588.002	Tool	Resource Development	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]
T1078	Valid Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	AA21-200A (APT40, 19 Jul 2021); AA24-038A (Volt Typhoon, 7 Feb 2024)	[29][33]
T1078.002	Domain Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	AA24-190A (APT40, 8 Jul 2024)	[37]
T1133	External Remote Services	Persistence, Initial Access	AA21-200A (APT40, 19 Jul 2021); AA22-158A (PRC actors, network devices, 7 Jun 2022); AA24-038A (Volt Typhoon, 7 Feb 2024)	[29][97][33]
T1189	Drive-by Compromise	Initial Access	AA21-200A (APT40, 19 Jul 2021)	[29]

Id	Name	Tactic (v20.0)	Where it appears in the record	Source
T1190	Exploit Public-Facing Application	Initial Access	AA21-200A (APT40, 19 Jul 2021); AA22-158A (PRC actors, network devices, 7 Jun 2022); AA23-144A (Volt Typhoon, 24 May 2023); AA24-038A (Volt Typhoon, 7 Feb 2024); AA24-190A (APT40, 8 Jul 2024); AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[29][97][31][33][37][53]
T1199	Trusted Relationship	Initial Access	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]
T1566.001	Spearphishing Attachment	Initial Access	AA21-200A (APT40, 19 Jul 2021)	[29]
T1566.002	Spearphishing Link	Initial Access	AA21-200A (APT40, 19 Jul 2021)	[29]
T1047	Windows Management Instrumentation	Execution	AA23-144A (Volt Typhoon, 24 May 2023); AA24-038A (Volt Typhoon, 7 Feb 2024)	[31][33]
T1053.005	Scheduled Task	Execution, Persistence, Privilege Escalation	AA24-190A (APT40, 8 Jul 2024)	[37]
T1059.001	PowerShell	Execution	AA23-144A (Volt Typhoon, 24 May 2023); AA24-038A (Volt Typhoon, 7 Feb 2024)	[31][33]
T1059.003	Windows Command Shell	Execution	AA23-144A (Volt Typhoon, 24 May 2023)	[31]
T1059.008	Network Device CLI	Execution	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]
T1072	Software Deployment Tools	Execution, Lateral Movement	AA24-190A (APT40, 8 Jul 2024)	[37]
T1203	Exploitation for Client Execution	Execution	AA21-200A (APT40, 19 Jul 2021); AA24-190A (APT40, 8 Jul 2024)	[29][37]
T1204.002	Malicious File	Execution	AA21-200A (APT40, 19 Jul 2021); AA24-190A (APT40, 8 Jul 2024)	[29][37]
T1574.001	DLL	Stealth, Execution	AA24-190A (APT40, 8 Jul 2024)	[37]
T1609	Container Administration Command	Execution	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]
T1610	Deploy Container	Execution	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]
T1098.004	SSH Authorized Keys	Persistence, Privilege Escalation	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]
T1505.003	Web Shell	Persistence	AA21-200A (APT40, 19 Jul 2021); AA23-144A (Volt Typhoon, 24 May 2023); AA24-190A (APT40, 8 Jul 2024)	[29][31][37]
T1543.005	Container Service	Persistence, Privilege Escalation	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]

Id	Name	Tactic (v20.0)	Where it appears in the record	Source
T1547.001	Registry Run Keys / Startup Folder	Persistence, Privilege Escalation	AA24-190A (APT40, 8 Jul 2024)	[37]
T1556	Modify Authentication Process	Defense Impairment, Persistence, Credential Access	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]
T1006	Direct Volume Access	Stealth	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1036.005	Match Legitimate Resource Name or Location	Stealth	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1070.004	File Deletion	Stealth	AA24-038A (Volt Typhoon, 7 Feb 2024); AA24-190A (APT40, 8 Jul 2024)	[33][37]
T1070.009	Clear Persistence	Stealth	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1140	Deobfuscate/Decode Files or Information	Stealth	AA24-190A (APT40, 8 Jul 2024)	[37]
T1218	System Binary Proxy Execution	Stealth	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1599	Network Boundary Bridging	Defense Impairment	AA22-158A (PRC actors, network devices, 7 Jun 2022); AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[97][53]
T1685.005	Clear Windows Event Logs (reported as T1070.001, revoked in v20.0)	Defense Impairment	AA23-144A (Volt Typhoon, 24 May 2023); AA24-038A (Volt Typhoon, 7 Feb 2024)	[31][33][4]
T1686	Disable or Modify System Firewall (reported as T1562.004, revoked in v20.0)	Defense Impairment	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53][4]
T1003.001	LSASS Memory	Credential Access	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1003.003	NTDS	Credential Access	AA23-144A (Volt Typhoon, 24 May 2023); AA24-038A (Volt Typhoon, 7 Feb 2024)	[31][33]
T1040	Network Sniffing	Credential Access, Discovery	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]
T1056.003	Web Portal Capture	Collection, Credential Access	AA24-190A (APT40, 8 Jul 2024)	[37]
T1110.003	Password Spraying	Credential Access	AA23-144A (Volt Typhoon, 24 May 2023)	[31]
T1111	Multi-Factor Authentication Interception	Credential Access	AA24-190A (APT40, 8 Jul 2024)	[37]
T1528	Steal Application Access Token	Credential Access	AA24-190A (APT40, 8 Jul 2024)	[37]

Id	Name	Tactic (v20.0)	Where it appears in the record	Source
T1539	Steal Web Session Cookie	Credential Access	AA24-190A (APT40, 8 Jul 2024)	[37]
T1555	Credentials from Password Stores	Credential Access	AA23-144A (Volt Typhoon, 24 May 2023)	[31]
T1558.003	Kerberoasting	Credential Access	AA24-190A (APT40, 8 Jul 2024)	[37]
T1016	System Network Configuration Discovery	Discovery	AA23-144A (Volt Typhoon, 24 May 2023); AA24-038A (Volt Typhoon, 7 Feb 2024); AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[31][33][53]
T1018	Remote System Discovery	Discovery	AA24-190A (APT40, 8 Jul 2024)	[37]
T1033	System Owner/User Discovery	Discovery	AA23-144A (Volt Typhoon, 24 May 2023); AA24-038A (Volt Typhoon, 7 Feb 2024); AA24-190A (APT40, 8 Jul 2024)	[31][33][37]
T1046	Network Service Discovery	Discovery	AA24-038A (Volt Typhoon, 7 Feb 2024); AA24-190A (APT40, 8 Jul 2024)	[33][37]
T1049	System Network Connections Discovery	Discovery	AA24-190A (APT40, 8 Jul 2024)	[37]
T1057	Process Discovery	Discovery	AA24-038A (Volt Typhoon, 7 Feb 2024); AA24-190A (APT40, 8 Jul 2024)	[33][37]
T1069.001	Local Groups	Discovery	AA23-144A (Volt Typhoon, 24 May 2023)	[31]
T1069.002	Domain Groups	Discovery	AA23-144A (Volt Typhoon, 24 May 2023)	[31]
T1082	System Information Discovery	Discovery	AA23-144A (Volt Typhoon, 24 May 2023); AA24-038A (Volt Typhoon, 7 Feb 2024); AA24-190A (APT40, 8 Jul 2024); AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[31][33][37][53]
T1083	File and Directory Discovery	Discovery	AA24-038A (Volt Typhoon, 7 Feb 2024); AA24-190A (APT40, 8 Jul 2024)	[33][37]
T1654	Log Enumeration	Discovery	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1021.001	Remote Desktop Protocol	Lateral Movement	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1021.002	SMB/Windows Admin Shares	Lateral Movement	AA24-190A (APT40, 8 Jul 2024)	[37]
T1021.007	Cloud Services	Lateral Movement	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1563.002	RDP Hijacking	Lateral Movement	AA24-190A (APT40, 8 Jul 2024)	[37]

Id	Name	Tactic (v20.0)	Where it appears in the record	Source
T1005	Data from Local System	Collection	AA24-190A (APT40, 8 Jul 2024); AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[37][53]
T1560	Archive Collected Data	Collection	AA21-200A (APT40, 19 Jul 2021)	[29]
T1560.001	Archive via Utility	Collection	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1602.001	SNMP (MIB Dump)	Collection	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]
T1602.002	Network Device Configuration Dump	Collection	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]
T1001.003	Protocol or Service Impersonation	Command and Control	AA24-190A (APT40, 8 Jul 2024)	[37]
T1071.001	Web Protocols	Command and Control	AA24-190A (APT40, 8 Jul 2024)	[37]
T1090.001	Internal Proxy	Command and Control	AA24-038A (Volt Typhoon, 7 Feb 2024)	[33]
T1090.002	External Proxy	Command and Control	AA23-144A (Volt Typhoon, 24 May 2023)	[31]
T1090.003	Multi-hop Proxy	Command and Control	AA21-200A (APT40, 19 Jul 2021); AA24-038A (Volt Typhoon, 7 Feb 2024); AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[29][33][53]
T1105	Ingress Tool Transfer	Command and Control	AA24-038A (Volt Typhoon, 7 Feb 2024); AA24-190A (APT40, 8 Jul 2024)	[33][37]
T1572	Protocol Tunneling	Command and Control	AA21-200A (APT40, 19 Jul 2021); AA22-158A (PRC actors, network devices, 7 Jun 2022); AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[29][97][53]
T1020.001	Traffic Duplication	Exfiltration	AA22-158A (PRC actors, network devices, 7 Jun 2022)	[97]
T1041	Exfiltration Over C2 Channel	Exfiltration	AA21-200A (APT40, 19 Jul 2021); AA24-190A (APT40, 8 Jul 2024)	[29][37]
T1048.003	Exfiltration Over Unencrypted Non-C2 Protocol	Exfiltration	AA25-239A (Salt Typhoon cluster, 27 Aug 2025)	[53]

ATT&CK Enterprise techniques most shared across China-attributed groups

ATT&CK v20.0 records 320 distinct Enterprise techniques across the 46 groups (Scarlet Mimic set aside); this table gives the 30 most shared, with the number of groups and whether a verified advisory names the same id (Confirmed [3]; names and tactics [101]). All 30 are named in at least one of the six advisories and appear in the table above (Confirmed [29][31][33][37][53]). The counts reflect how much has been reported about each group, not how often each technique is used (Inferred [3]).

Rank	Id	Name	Tactic (v20.0)	Groups	In an advisory above
1	T1105	Ingress Tool Transfer	Command and Control	23	yes
2	T1059.001	PowerShell	Execution	21	yes
3	T1588.002	Tool	Resource Development	21	yes
4	T1059.003	Windows Command Shell	Execution	20	yes
5	T1204.002	Malicious File	Execution	19	yes
6	T1566.001	Spearphishing Attachment	Initial Access	19	yes
7	T1016	System Network Configuration Discovery	Discovery	18	yes
8	T1574.001	DLL	Stealth, Execution	18	yes
9	T1018	Remote System Discovery	Discovery	17	yes
10	T1190	Exploit Public-Facing Application	Initial Access	17	yes
11	T1003.001	LSASS Memory	Credential Access	16	yes
12	T1083	File and Directory Discovery	Discovery	16	yes
13	T1560.001	Archive via Utility	Collection	16	yes
14	T1005	Data from Local System	Collection	15	yes
15	T1036.005	Match Legitimate Resource Name or Location	Stealth	15	yes
16	T1203	Exploitation for Client Execution	Execution	15	yes
17	T1033	System Owner/User Discovery	Discovery	14	yes
18	T1047	Windows Management Instrumentation	Execution	14	yes
19	T1057	Process Discovery	Discovery	14	yes
20	T1078	Valid Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	14	yes
21	T1049	System Network Connections Discovery	Discovery	13	yes
22	T1070.004	File Deletion	Stealth	13	yes
23	T1071.001	Web Protocols	Command and Control	13	yes
24	T1547.001	Registry Run Keys / Startup Folder	Persistence, Privilege Escalation	13	yes
25	T1021.001	Remote Desktop Protocol	Lateral Movement	12	yes
26	T1082	System Information Discovery	Discovery	12	yes
27	T1140	Deobfuscate/Decode Files or Information	Stealth	12	yes
28	T1046	Network Service Discovery	Discovery	11	yes

Rank	Id	Name	Tactic (v20.0)	Groups	In an advisory above
29	T1053.005	Scheduled Task	Execution, Persistence, Privilege Escalation	11	yes
30	T1566.002	Spearphishing Link	Initial Access	11	yes

In the 15 groups this study profiles most closely (G0004, G0006, G0022, G0027, G0045, G0065, G0096, G0125, G0128, G0129, G0143, G1017, G1023, G1045, G1048) the leaders are T1059.003 (13 groups), T1003.001 (12), T1105 (12), T1016 (11), T1560.001 (11), T1059.001 (11), T1005 (10) and T1190 (10), across 286 distinct techniques (Confirmed [3]). Spearphishing Attachment is recorded for 19 of the 46 groups in the full set but for only 6 of these 15, while T1190 holds at 10 of 15: the groups with recent advisories lean on edge and public-facing exploitation (Inferred [3]).

ATT&CK groups

Group	Id and library page	Government or vendor record in this study	Source
APT1	G0006	Unit 61398 indictment, 2014; Mandiant, 2013	[18][55]
menuPass (APT10)	G0045	Indictment and UK attribution, 2018; BAE, 2017	[21][22][58]
Leviathan (APT40)	G0065	Indictment, AA21-200A, 2021; AA24-190A, 2024	[28][29][37]
ZIRCONIUM (APT31)	G0128	Indictment, OFAC, UK, 2024; Czech attribution, 2025	[34][35][36][50]
Threat Group-3390 (APT27)	G0027	i-Soon and APT27 charges, OFAC, 2025	[47][49]
HAFNIUM (Silk Typhoon)	G0125	Microsoft, 2021 and 2025; US, UK attribution 2021; Xu Zewei, 2025	[59][60][30][52][69]
APT41	G0096	Charges, 2020; home study	[27]
Volt Typhoon	G1017	AA23-144A, AA24-038A, KV Botnet disruption	[31][32][33]
Salt Typhoon	G1045	FBI and CISA statement, OFAC, AA25-239A	[40][46][53]
UNC3886	G1048	Mandiant, 2025	[71]
Ke3chang (NICKEL)	G0004	Microsoft DCU seizure, 2021	[61]
APT3	G0022	ATT&CK: attributed by researchers to the MSS [1]; Boyusec charges, 2017, do not name APT3 [19]	[1][19]
Deep Panda	G0009	Anthem per ATT&CK; Anthem indictment names no group	[1][23]
Aquatic Panda	G0143	i-Soon industry name, FBI 2025	[48][96]
MirrorFace	G1054	Japan NPA and NISC, 2025	[45]
Mustang Panda	G0129	ATT&CK record	[8]
Chimera	G0114	ATT&CK record	[11]
APT5	G1023	ATT&CK record (SPACEHOP)	[1][17]
PittyTiger	G0011	ATT&CK record; HC3 equates it with APT24, not bound here	[1][83]
APT18	G0026	HC3; no China wording in ATT&CK	[83][86][1]

Actors in the record with no ATT&CK group object in the library: Flax Typhoon, Storm-0558, UNC5221 (ARMORY CASTLE), UNC2814, UNC6508, Storm-2603, APT22, APT20, APT9 and APT24 (Confirmed [1][6]). Linen Typhoon and Violet Typhoon are aliases of G0027 and G0128 (Confirmed [1]). All other groups in Part 1 have a page under /mitre/groups/.

ATT&CK campaigns

Campaign	Id and library page	Dates	Group	Source
APT41 campaign	C0017	May 2021 to Feb 2022	G0096	[9]
APT41 DUST	C0040	Jan 2023 to Jun 2024	G0096	[9]
KV Botnet Activity	C0035	Oct 2022 to Jan 2024	G1017	[10]
Versa Director Zero Day Exploitation	C0039	Jun to Aug 2024	G1017	[10]
Leviathan Australian Intrusions	C0049	Apr to Sep 2022	G0065	[14]
RedDelta Modified PlugX Infection Chain Operations	C0047	Jul 2023 to Dec 2024	G0129	[8]
SPACEHOP Activity	C0052	Jan 2019 to May 2024	G1023, G0004	[17]
RedPenguin	C0056	Jul 2024 to Mar 2025	G1048	[15]
Operation AkaiRyu	C0060	Jun to Sep 2024	G1054 (ATT&CK places MirrorFace under the menuPass umbrella; no G0045 link on the campaign)	[1]
Indian Critical Infrastructure Intrusions	C0043	Jan 2021 to Apr 2022	none attached; the description names RedEcho (G1042) and TAG38	[2]
SharePoint ToolShell Exploitation	C0058	Jul 2025	G0027, G0128	[2]
Night Dragon, FunnyDream, CuckooBees, Wocao, Cutting Edge, FLORAHOX, Quad7, Digital Eye, AI-orchestrated campaign	C0002, C0007, C0012, C0014, C0029, C0053, C0055, C0061, C0062	See Part 1	No group object (C0053 is linked to ZIRCONIUM in the library data)	[2]

ATT&CK software

The shared-software table in Part 6 links each id. Additional software bound by the record: S1206 JumbledPath (Salt Typhoon), S1144 FRP, S1154 VersaMem, S0108 netsh, S0359 Nltest and S0645 Wevtutil (Volt Typhoon) (Confirmed [10][12]), and S9015 BRICKSTORM (UNC5221, BRICKSTORM [70][73]; no group attached in ATT&CK, Confirmed [3]).

CAPEC and CWE

No advisory or vendor report in this study names a CAPEC pattern. The CAPEC entries below are the library's own technique-to-CAPEC mappings in `technique_profiles.jsonl` for the advisory techniques in this Part, limited to those that describe a behavior the record shows (Confirmed [101] for the mapping; the selection is Inferred). Other library mappings for these techniques (for example CAPEC-665 for T1556, CAPEC-187 and CAPEC-657 for T1072) describe mechanisms not in the record and are left out.

CAPEC	Name	Technique it maps to	Where the behavior appears
CAPEC-163	Spear Phishing	T1566.001, T1566.002	AA21-200A [29]
CAPEC-555	Remote Services with Stolen Credentials	T1133	AA21-200A, AA22-158A, AA24-038A [29][97][33]
CAPEC-560	Use of Known Domain Credentials	T1078	AA21-200A, AA24-038A [29][33]
CAPEC-650	Upload a Web Shell to a Web Server	T1505.003	AA21-200A, AA23-144A, AA24-190A [29][31][37]
CAPEC-565	Password Spraying	T1110.003	AA23-144A [31]
CAPEC-509	Kerberoasting	T1558.003	AA24-190A [37]
CAPEC-561	Windows Admin Shares with Stolen Credentials	T1021.002	AA24-190A [37]
CAPEC-31	Accessing/Intercepting/Modifying HTTP Cookies	T1539	AA24-190A [37]
CAPEC-158	Sniffing Network Traffic	T1040	AA25-239A [53]
CAPEC-700	Network Boundary Bridging	T1599	AA22-158A, AA25-239A [97][53]

CWE: the weakness class of each named CVE is set by its NVD record, and those records were not opened for this study, so no CWE ids are bound here (not checked). The CVE-level CWE for Log4Shell is in the Log4Shell study.

D3FEND countermeasures

Three advisories map their mitigations to D3FEND: AA25-239A (Appendix C, D3FEND v1.2.0), AA22-158A and AA21-200B (Confirmed [53][97][93]). Together they name 55 distinct countermeasures, counting the D3-CFP that AA25-239A's rendered page prints for changing default passwords as the library's D3-CDP, and the one D3-SPA that AA21-200B prints against Process Spawn Analysis as D3-PSA, the id it uses elsewhere (Confirmed [53][93]). Forty of the 55 have a library page, and the table lists all 40: the ten named in AA25-239A or AA22-158A first, then the 30 that only AA21-200B names, which include D3-CSPP, D3-DA, D3-DQSA, D3-FA, D3-FC, D3-HD, D3-MENCR, D3-PHDURA, D3-PMAD, D3-SBV, D3-SCA, D3-SDM, D3-SFA, D3-SJA, D3-SMRA, D3-SRA and D3-UA. Each row gives the techniques in this Part's advisory table that the library's `d3fend_countermeasures.jsonl` maps to it, where there are any (Confirmed [5]). The other fifteen are absent from the library's D3FEND 1.6.0 data, so they are not linked: D3-PM Platform Monitoring, D3-NI Network Isolation, D3-NTA Network Traffic Analysis, D3-FCR, D3-PA, D3-SEA, D3-OSM, D3-FAPA, D3-UBA, D3-ANET, D3-JFAPA, D3-AH, D3-PH, D3-TAAN and D3-MAC (Confirmed [5]). AA21-200B itself cautions that its D3FEND mapping may not match automated mappings (Confirmed [93]).

Countermeasure	Id	D3FEND tactic	Named in	Library-mapped techniques from this study	Source
Network Traffic Community Deviation	D3-NTCD	Detect	AA25-239A	18, including T1190, T1199, T1572, T1090.003, T1048.003	[53][5]
Outbound Traffic Filtering	D3-OTF	Isolate	AA25-239A, AA22-158A, AA21-200B	T1048.003, T1071.001, T1090.002, T1090.003, T1105, T1572	[53][97][93][5]
Application Configuration Hardening	D3-ACH	Harden	AA25-239A, AA22-158A	none in this set	[53][97][5]
Change Default Password	D3-CDP	Harden	AA25-239A (the advisory's rendered page shows D3-CFP; the library id is D3-CDP)	T1078, T1078.002, T1110.003	[53][5]
Credential Hardening	D3-CH	Harden	AA25-239A, AA21-200B	T1003.003, T1110.003, T1528, T1539	[53][93][5]
Software Update	D3-SU	Harden	AA25-239A, AA22-158A, AA21-200B	T1056.003, T1072	[53][97][93][5]
Inbound Traffic Filtering	D3-ITF	Isolate	AA25-239A, AA22-158A, AA21-200B	T1190, T1566.001, T1566.002	[53][97][93][5]
Multi-factor Authentication	D3-MFA	Harden	AA22-158A, AA21-200B	T1003.003, T1110.003, T1528, T1539	[97][93][5]
Strong Password Policy	D3-SPP	Harden	AA22-158A	T1110.003	[97][5]
User Account Permissions	D3-UAP	Isolate	AA22-158A	T1078, T1078.002	[97][5]
Connection Attempt Analysis	D3-CAA	Detect	AA21-200B	T1047, T1090.001, T1110.003, T1199	[93][5]

Countermeasure	Id	D3FEND tactic	Named in	Library-mapped techniques from this study	Source
Process Lineage Analysis	D3-PLA	Detect	AA21-200B	T1003.001, T1033, T1053.005, T1505.003, T1556	[93][5]
Process Spawn Analysis	D3-PSA	Detect	AA21-200B	T1003.001, T1016, T1018, T1033, T1047, T1053.005, T1057, T1082, T1140, T1505.003, T1556	[93][5]
Administrative Network Activity Analysis	D3-ANAA	Detect	AA21-200B	T1047, T1110.003	[93][5]
Relay Pattern Analysis	D3-RPA	Detect	AA21-200B	T1048.003, T1071.001, T1090.002, T1090.003, T1105, T1572	[93][5]
User Geolocation Logon Pattern Analysis	D3-UGLPA	Detect	AA21-200B	18, the same set as D3-NTCD	[93][5]
Remote Terminal Session Detection	D3-RTSD	Detect	AA21-200B	18, the same set as D3-NTCD	[93][5]
DNS Traffic Analysis	D3-DNSTA	Detect	AA21-200B	T1040	[93][5]
Message Authentication	D3-MAN	Harden	AA21-200B	none in this set	[93][5]
Executable Allowlisting	D3-EAL	Isolate	AA21-200B	T1016, T1018, T1033, T1047, T1057, T1082, T1140, T1204.002, T1505.003, T1547.001	[93][5]
Executable Denylisting	D3-EDL	Isolate	AA21-200B	T1016, T1018, T1033, T1047, T1057, T1082, T1140, T1204.002, T1505.003, T1547.001	[93][5]
DNS Denylisting	D3-DNSDL	Isolate	AA21-200B	none in this set	[93][5]
Hardware-based Process Isolation	D3-HBPI	Isolate	AA21-200B	T1003.001, T1016, T1018, T1033, T1047, T1053.005, T1057, T1082, T1140, T1505.003, T1556	[93][5]
Client-server Payload Profiling	D3-CSPP	Detect	AA21-200B	18, the same set as D3-NTCD	[93][5]
Dynamic Analysis	D3-DA	Detect	AA21-200B	T1016, T1140, T1204.002, T1505.003, T1547.001, T1566.001, T1566.002	[93][5]
Database Query String Analysis	D3-DQSA	Detect	AA21-200B	T1190	[93][5]
File Analysis	D3-FA	Detect	AA21-200B	T1005, T1016, T1018, T1033, T1036.005, T1041, T1070.004, T1071.001, T1072, T1083, T1140, T1204.002, T1505.003, T1547.001, T1555, T1560, T1560.001, T1566.001, T1566.002, T1574.001	[93][5]
File Carving	D3-FC	Detect	AA21-200B	none in this set	[93][5]

Countermeasure	Id	D3FEND tactic	Named in	Library-mapped techniques from this study	Source
Homoglyph Detection	D3-HD	Detect	AA21-200B	T1189, T1566.001, T1566.002	[93][5]
Message Encryption	D3-MENCR	Harden	AA21-200B	none in this set	[93][5]
Per Host Download-Upload Ratio Analysis	D3-PHDURA	Detect	AA21-200B	18, the same set as D3-NTCD	[93][5]
Protocol Metadata Anomaly Detection	D3-PMAD	Detect	AA21-200B	18, the same set as D3-NTCD	[93][5]
Service Binary Verification	D3-SBV	Detect	AA21-200B	T1056.003, T1072	[93][5]
System Call Analysis	D3-SCA	Detect	AA21-200B	T1016, T1018, T1033, T1036.005, T1047, T1049, T1057, T1082, T1140	[93][5]
System Daemon Monitoring	D3-SDM	Detect	AA21-200B	T1053.005	[93][5]
System File Analysis	D3-SFA	Detect	AA21-200B	T1018	[93][5]
Scheduled Job Analysis	D3-SJA	Detect	AA21-200B	none in this set	[93][5]
Sender MTA Reputation Analysis	D3-SMRA	Detect	AA21-200B	T1566.001, T1566.002	[93][5]
Sender Reputation Analysis	D3-SRA	Detect	AA21-200B	T1566.001, T1566.002	[93][5]
URL Analysis	D3-UA	Detect	AA21-200B	T1189, T1566.002	[93][5]

A note on the automated join. Joining the 30 most shared techniques to the library's `technique_to_d3fend.jsonl` is dominated by generic file-artifact countermeasures (Content Filtering, File Analysis and File Integrity Monitoring each cover 15 of the 30), and four of the 30 (T1588.002, T1059.001, T1059.003, T1046) have no mapping at all; the advisory-named countermeasures above are the better guide, and the automated join is secondary (Inferred [5]). The six countermeasures that map to the same 18 techniques (D3-NTCD, D3-UGLPA, D3-RTSD, D3-CSPP, D3-PHDURA, D3-PMAD) show the same effect: they map to network-facing techniques broadly rather than to these specific advisories (Inferred [5]).

Navigator layer

The study's techniques are saved as an ATT&CK Navigator layer, navigator/library-layers/threat-actors-china.json, listed on the Navigator layers page. It is pinned to ATT&CK v20 and holds the 77 techniques of the advisory table in this Part, each named in at least one of the six government advisories and scored 100; all 30 of the most shared techniques are among them. Each entry's comment names the source numbers above.

Cross-cutting themes

- **The state works through companies.** Front companies (Hainan Xiandun, Wuhan Xiaoruizhi), contractors (Huaying Haitai, Shanghai Powerock, i-Soon, Sichuan Silence) and infrastructure vendors (Integrity Tech, Sichuan Juxinhe) recur across a decade of indictments and sanctions (Confirmed [21][28][35][52][47][42][44][46]). Sanctions now target the companies directly, a lever that does not depend on a defendant travelling to a country that will extradite, as Xu Zewei did (Inferred from [52][92]).
- **The edge replaced the inbox.** From spearphishing and watering holes in AA21-200A to network-device CLI, container services on routers and configuration dumps in AA25-239A, the advisories trace the move to devices that most security tools do not watch (Confirmed [29][53]; the trend reading is Inferred).
- **Infrastructure is borrowed or bought, not built.** KV Botnet, the Integrity Tech botnet, Raptor Train (over 200,000 devices in 2024) and the covert networks NCSC describes mean an intrusion's source address says little about who is behind it (Confirmed [32][38][76][74]; the consequence is Inferred).
- **Names are a mess, and the sources disagree on the record.** Silk Typhoon sits under HAFNIUM in ATT&CK and Microsoft but under APT27 actors in the FBI's PSA; GTIG says UNC5221 is not Silk Typhoon while Microsoft lists for Silk Typhoon CVE-2025-0282, which Mandiant attributes to UNC5221; AA25-239A says the Salt Typhoon vendor names partially overlap and are not one to one (Confirmed [1][48][68][69][70][53]). Map names to ids, and carry the source with every mapping (Inferred). See Threat Actors.
- **Healthcare is an intelligence target first.** The record is research theft, identity data and mail, with ransomware capability present in the ecosystem but no state-attributed disruptive hospital attack in these sources (Inferred from Part 5).

Where the sources disagree

Question	One source	Another source
When Volt Typhoon started	Microsoft: mid-2021 [62]; ATT&CK: since at least 2021 [10]	AA24-038A: footholds of at least five years in some victims [33]
Volt Typhoon's intent	Microsoft: moderate confidence on developing disruption capability [62]	AA24-038A: high confidence of pre-positioning for OT disruption [33]
When Salt Typhoon started	Treasury and ATT&CK: since at least 2019 [46][12]	AA25-239A: since at least 2021 [53]
Who Silk Typhoon is	ATT&CK and Microsoft: HAFNIUM [1][69]	FBI PSA: a name for APT27 actors Yin and Zhou [48]
CVE-2025-0282	Microsoft: in Silk Typhoon's zero-days [69]	Mandiant and GTIG: UNC5221, which is not Silk Typhoon [68][70]
BRICKSTORM	GTIG: suspected China-nexus [70]	CISA, NSA and the Cyber Centre: PRC state-sponsored [73]
APT40's Hainan basing	AA21-200A: stated [29]	AA24-190A: cited as prior reporting [37]
APT10 and healthcare	DOJ release: healthcare among target industries [21]	FBI wanted notice: healthcare not listed [90]
OPM 2015	Not checked: this study opened no source that names China for OPM	House Oversight report: "a foreign adversary" only [57]
Who Charcoal Typhoon is	ATT&CK: alias of Earth Lusca G1006 [1]	FBI PSA: an industry name for i-Soon [48]
GRIDTIDE reach	GTIG body: 53 confirmed victims in 42 countries, plus 20 more countries	The same post's conclusion: over 70 countries [75]
Volt Typhoon itself	Ten agencies in five countries [33]	China's CVERC: a US false flag (Reported) [63]

What is overstated and what is understated

- **Overstated: "Chinese hackers" as one actor.** The record names separate PLA, MSS and MPS lines, provincial bureaus, contractors and brokers with different missions (Confirmed [18][21][35][42][47]). Treating them as one blurs which defenses matter (Inferred).
- **Overstated: novelty.** The telecom guidance found that exploitation matched existing weaknesses and observed no novel activity, and AA25-239A observed no zero-day use (Confirmed [41][53]).
- **Understated: dwell time.** Five-year footholds and a 393-day average dwell mean a clean scan today says little about the last year (Confirmed [33][70]; the reading is Inferred).
- **Understated: health research as a target.** The healthcare focus usually falls on ransomware (Inferred); the state-attributed record is research and identity data theft from 2011 to 2025 (Confirmed [87][23][78][80]).

Sources

All sources were opened on 3 November 2026. "Primary" means the organization that published the fact; grades apply to the source as used here. The library data files are at commit e85b559 or current main.

#	Source	URL	Opened	Grade
1	TeamStarWolf library, ATT&CK v20.0 groups data (`groups.jsonl`)	github.com/TeamStarWolf/TeamStarWolf/.../data/attack/groups.jsonl	2026-11-03	Confirmed
2	TeamStarWolf library, campaign data (`campaign_profiles.jsonl`, `campaigns.jsonl`)	github.com/TeamStarWolf/TeamStarWolf/.../data/attack/campaign_profiles.jsonl	2026-11-03	Confirmed
3	TeamStarWolf library, group technique and software data (`group_to_technique.jsonl`, `group_profiles.jsonl`)	github.com/TeamStarWolf/TeamStarWolf/.../data/attack/group_to_technique.jsonl	2026-11-03	Confirmed
4	TeamStarWolf library, revoked ATT&CK ids (`superseded_by.jsonl`)	github.com/TeamStarWolf/TeamStarWolf/.../data/attack/superseded_by.jsonl	2026-11-03	Confirmed
5	TeamStarWolf library, D3FEND data (`d3fend_countermeasures.jsonl`, `technique_to_d3fend.jsonl`)	github.com/TeamStarWolf/TeamStarWolf/.../data/attack/d3fend_countermeasures.jsonl	2026-11-03	Confirmed
6	TeamStarWolf library, generated group pages (`mitre/groups/`)	github.com/TeamStarWolf/TeamStarWolf/tree/main/mitre/groups	2026-11-03	Confirmed
7	TeamStarWolf library, ATT&CK data tree including ICS and Mobile	github.com/TeamStarWolf/TeamStarWolf/tree/.../data/attack	2026-11-03	Confirmed
8	MITRE ATT&CK, G0129 Mustang Panda	attack.mitre.org/groups/G0129	2026-11-03	Confirmed
9	MITRE ATT&CK, G0096 APT41	attack.mitre.org/groups/G0096	2026-11-03	Confirmed
10	MITRE ATT&CK, G1017 Volt Typhoon	attack.mitre.org/groups/G1017	2026-11-03	Confirmed
11	MITRE ATT&CK, G0114 Chimera	attack.mitre.org/groups/G0114	2026-11-03	Confirmed
12	MITRE ATT&CK, G1045 Salt Typhoon	attack.mitre.org/groups/G1045	2026-11-03	Confirmed
13	MITRE ATT&CK, G0027 Threat Group-3390	attack.mitre.org/groups/G0027	2026-11-03	Confirmed
14	MITRE ATT&CK, G0065 Leviathan	attack.mitre.org/groups/G0065	2026-11-03	Confirmed
15	MITRE ATT&CK, G1048 UNC3886	attack.mitre.org/groups/G1048	2026-11-03	Confirmed
16	MITRE ATT&CK, G0045 menuPass	attack.mitre.org/groups/G0045	2026-11-03	Confirmed
17	MITRE ATT&CK, G0004 Ke3chang	attack.mitre.org/groups/G0004	2026-11-03	Confirmed
18	US DOJ, five Chinese military hackers charged, 19 May 2014	justice.gov/opa/pr/us-charges-five-chinese-military-hackers...	2026-11-03	Confirmed
19	US DOJ, three Boyusec employees charged, 27 Nov 2017	justice.gov/archives/opa/pr/us-charges-three-chinese-hackers...	2026-11-03	Confirmed
20	US DOJ, Jiangsu MSS officers and recruits charged, 30 Oct 2018	justice.gov/archives/opa/pr/chinese-intelligence-officers...	2026-11-03	Confirmed
21	US DOJ, two APT10 hackers charged, 20 Dec 2018	justice.gov/archives/opa/pr/two-chinese-hackers-associated-ministry-state-security...	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
22	UK Government, UK and allies reveal global scale of Chinese cyber campaign, 20 Dec 2018	gov.uk/government/news/uk-and-allies-reveal-global-scale-of-chinese-cyber-campaign	2026-11-03	Confirmed
23	US DOJ, Anthem-era indictment of a China-based hacking group member, 9 May 2019	justice.gov/archives/opa/pr/member-sophisticated-china-based-hacking-group...	2026-11-03	Confirmed
24	US DOJ, PLA members charged over Equifax, 10 Feb 2020	justice.gov/archives/opa/pr/chinese-military-personnel-charged...	2026-11-03	Confirmed
25	CISA, FBI and CISA warn against Chinese targeting of COVID-19 research, 13 May 2020	cisa.gov/news-events/news/fbi-and-cisa-warn-against-chinese-targeting...	2026-11-03	Confirmed
26	US DOJ, Li Xiaoyu and Dong Jiazhi charged, 21 Jul 2020	justice.gov/archives/opa/pr/two-chinese-hackers-working-ministry-state-security...	2026-11-03	Confirmed
27	US DOJ, seven defendants including APT41 actors charged, 16 Sep 2020	justice.gov/archives/opa/pr/seven-international-cyber-defendants...	2026-11-03	Confirmed
28	US DOJ, four APT40 nationals charged, 19 Jul 2021	justice.gov/archives/opa/pr/four-chinese-nationals-working-ministry-state-security...	2026-11-03	Confirmed
29	CISA and FBI, AA21-200A, APT40 TTPs, 19 Jul 2021	cisa.gov/news-events/cybersecurity-advisories/aa21-200a	2026-11-03	Confirmed
30	UK Government, UK and allies hold Chinese state responsible, 19 Jul 2021	gov.uk/government/news/uk-and-allies-hold-chinese-state-responsible...	2026-11-03	Confirmed
31	NSA, CISA, FBI and partners, AA23-144A, Volt Typhoon, 24 May 2023	cisa.gov/news-events/cybersecurity-advisories/aa23-144a	2026-11-03	Confirmed
32	US DOJ, KV Botnet disruption, 31 Jan 2024	justice.gov/archives/opa/pr/us-government-disrupts-botnet...	2026-11-03	Confirmed
33	CISA and partners, AA24-038A, Volt Typhoon pre-positioning, 7 Feb 2024	cisa.gov/news-events/cybersecurity-advisories/aa24-038a	2026-11-03	Confirmed
34	US DOJ, seven APT31-linked hackers charged, 25 Mar 2024	justice.gov/archives/opa/pr/seven-hackers-associated-chinese-government...	2026-11-03	Confirmed
35	US Treasury, OFAC sanctions on Wuhan Xiaoruizhi and APT31 actors, 25 Mar 2024	home.treasury.gov/news/press-releases/jy2205	2026-11-03	Confirmed
36	UK Government, China state-affiliated organisations and individuals held responsible, 25 Mar 2024	gov.uk/government/news/uk-holds-china-state-affiliated-organisations...	2026-11-03	Confirmed
37	ASD's ACSC and partners, AA24-190A, APT40, 8 Jul 2024	cisa.gov/news-events/cybersecurity-advisories/aa24-190a	2026-11-03	Confirmed
38	US DOJ, court-authorized disruption of the Integrity Tech botnet, 18 Sep 2024	justice.gov/archives/opa/pr/court-authorized-operation-disrupts-worldwide-botnet...	2026-11-03	Confirmed
39	FBI, CNMF and NSA, JCSA-20240918-001, Integrity Tech botnet	ic3.gov/CSA/2024/240918.pdf	2026-11-03	Confirmed
40	FBI and CISA, joint statement on PRC targeting of commercial telecommunications, 13 Nov 2024	cisa.gov/news-events/news/joint-statement-fbi-and-cisa...	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
41	CISA and partners, Enhanced Visibility and Hardening Guidance for Communications Infrastructure, 4 Dec 2024	cisa.gov/resources-tools/resources/enhanced-visibility-and-hardening-guidance...	2026-11-03	Confirmed
42	US DOJ, Guan Tianfeng charged, 10 Dec 2024	justice.gov/archives/opa/pr/china-based-hacker-charged...	2026-11-03	Confirmed
43	US Treasury, OFAC designates Sichuan Silence and Guan Tianfeng, 10 Dec 2024	home.treasury.gov/news/press-releases/jy2742	2026-11-03	Confirmed
44	US Treasury, OFAC designates Integrity Technology Group, 3 Jan 2025	home.treasury.gov/news/press-releases/jy2769	2026-11-03	Confirmed
45	Japan National Police Agency, MirrorFace caution, 8 Jan 2025	npa.go.jp/bureau/cyber/koho/caution/caution20250108.html	2026-11-03	Confirmed
46	US Treasury, OFAC sanctions Yin Kecheng and Sichuan Juxinhe, 17 Jan 2025	home.treasury.gov/news/press-releases/jy2792	2026-11-03	Confirmed
47	US DOJ, 12 Chinese contract hackers and law enforcement officers charged, 5 Mar 2025	justice.gov/opa/pr/justice-department-charges-12-chinese-contract-hackers...	2026-11-03	Confirmed
48	FBI, public service announcement PSA250305	ic3.gov/PSA/2025/PSA250305	2026-11-03	Confirmed
49	US Treasury, OFAC designates Zhou Shuai and Shanghai Heiying, 5 Mar 2025	home.treasury.gov/news/press-releases/sb0042	2026-11-03	Confirmed
50	Czech Ministry of Foreign Affairs, statement by the Government of the Czech Republic, 28 May 2025	mzv.gov.cz/.../statement_by_the_government_of_the_czech.html	2026-11-03	Confirmed
51	NUKIB, Czech government attributes cyberattacks to China, 28 May 2025	nukib.gov.cz/en/infoservis-en/news/2263-...	2026-11-03	Confirmed
52	US DOJ, arrest of a Chinese state-sponsored contract hacker (Xu Zewei), 8 Jul 2025	justice.gov/opa/pr/justice-department-announces-arrest-prolific-chinese...	2026-11-03	Confirmed
53	NSA, CISA, FBI, DC3 and partners, AA25-239A, Salt Typhoon cluster, 27 Aug 2025	cisa.gov/news-events/cybersecurity-advisories/aa25-239a	2026-11-03	Confirmed
54	Google, "A new approach to China", 12 Jan 2010	googleblog.blogspot.com/2010/01/new-approach-to-china.html	2026-11-03	Confirmed
55	Mandiant, APT1 report announcement, 19 Feb 2013	cloud.google.com/blog/topics/threat-intelligence/mandiant-exposes-apt1...	2026-11-03	Confirmed
56	Community Health Systems, Form 8-K, filed 18 Aug 2014	`https://www.sec.gov/Archives/edgar/data/1108109/000119312514312504/d776541d8k.htm`	2026-11-03	Confirmed
57	House Oversight Committee, OPM data breach majority report, 7 Sep 2016	oversight.house.gov/report/opm-data-breach...	2026-11-03	Confirmed
58	BAE Systems, APT10 Operation Cloud Hopper, 4 Apr 2017	`https://www.baesystems.com/en-uk/insight/apt10-operation-cloud-hopper`	2026-11-03	Confirmed
59	Microsoft, HAFNIUM targeting Exchange servers, 2 Mar 2021	microsoft.com/en-us/security/blog/2021/03/02/hafnium-targeting-exchange-servers	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
60	The White House (archived), US joined by allies attributes malicious cyber activity to the PRC, 19 Jul 2021	bidenwhitehouse.archives.gov/.../2021/07/19/...	2026-11-03	Confirmed
61	Microsoft, Digital Crimes Unit action against NICKEL, 6 Dec 2021	blogs.microsoft.com/on-the-issues/2021/12/06/cyberattacks-nickel-dcu-china	2026-11-03	Confirmed
62	Microsoft, Volt Typhoon targets US critical infrastructure, 24 May 2023	microsoft.com/en-us/security/blog/2023/05/24/volt-typhoon...	2026-11-03	Confirmed
63	CGTN (PRC state media), report on CVERC's third Volt Typhoon report, 14 Oct 2024	news.cgtn.com/news/2024-10-14/...	2026-11-03	Reported
64	Microsoft, analysis of Storm-0558 techniques, 14 Jul 2023	microsoft.com/en-us/security/blog/2023/07/14/analysis-of-storm-0558...	2026-11-03	Confirmed
65	NCSC, China state-affiliated actors target UK democratic institutions, 25 Mar 2024	ncsc.gov.uk/news/china-state-affiliated-actors-target-uk-democratic-institutions-parliamentarians	2026-11-03	Confirmed
66	NCSC, advice to counter a China-linked campaign targeting thousands of devices, 18 Sep 2024	ncsc.gov.uk/news/ncsc-and-partners-issue-advice...	2026-11-03	Confirmed
67	Congressional Research Service, IF12798, updated 23 Jan 2025	congress.gov/crs_external_products/IF/HTML/IF12798.web.html	2026-11-03	Confirmed
68	Mandiant, Ivanti Connect Secure VPN zero-day, 8 Jan 2025	cloud.google.com/blog/topics/threat-intelligence/ivanti-connect-secure-vpn-zero-day	2026-11-03	Confirmed
69	Microsoft, Silk Typhoon targeting IT supply chain, 5 Mar 2025	microsoft.com/en-us/security/blog/2025/03/05/silk-typhoon-targeting-it-supply-chain	2026-11-03	Confirmed
70	Google Threat Intelligence Group, BRICKSTORM espionage campaign, 24 Sep 2025	cloud.google.com/blog/topics/threat-intelligence/brickstorm-espionage-campaign	2026-11-03	Confirmed
71	Mandiant, China-nexus espionage targets Juniper routers, 11 Mar 2025	cloud.google.com/blog/topics/threat-intelligence/china-nexus-espionage-targets-juniper-routers	2026-11-03	Confirmed
72	Microsoft, disrupting active exploitation of on-premises SharePoint, 22 Jul 2025	microsoft.com/en-us/security/blog/2025/07/22/disrupting-active-exploitation...	2026-11-03	Confirmed
73	CISA, NSA and the Cyber Centre warn of BRICKSTORM, 4 Dec 2025	cisa.gov/news-events/news/cisa-nsa-and-cyber-centre-warn...	2026-11-03	Confirmed
74	NCSC, international cyber agencies' advice on China-linked covert networks, 23 Apr 2026	ncsc.gov.uk/news/international-cyber-agencies-fresh-advice...	2026-11-03	Confirmed
75	Google Threat Intelligence Group, disrupting the GRIDTIDE espionage campaign, 25 Feb 2026	cloud.google.com/blog/topics/threat-intelligence/disrupting-gridtide-global-espionage-campaign	2026-11-03	Confirmed
76	NCSC, defending against China-nexus covert networks of compromised devices, 23 Apr 2026	ncsc.gov.uk/news/defending-against-china-nexus-covert-networks...	2026-11-03	Confirmed
77	CISA, AA26-113A, China-nexus covert networks, 23 Apr 2026	cisa.gov/news-events/cybersecurity-advisories/aa26-113a	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
78	Google Threat Intelligence Group, PRC targets US medical research, 15 Jun 2026	cloud.google.com/blog/topics/threat-intelligence/prc-targets-us-medical-research	2026-11-03	Confirmed
79	NSA, CISA and FBI, AA26-251A, model distillation campaigns, 8 Sep 2026	cisa.gov/news-events/cybersecurity-advisories/aa26-251a	2026-11-03	Confirmed
80	CISA and partners, AA26-281A, actors enabled by Integrity Technology Group, 8 Oct 2026	cisa.gov/news-events/cybersecurity-advisories/aa26-281a	2026-11-03	Confirmed
81	NCSC, China-linked actors called out for targeting sensitive data, 8 Oct 2026	ncsc.gov.uk/news/china-linked-actors-called-out...	2026-11-03	Confirmed
82	Google Threat Intelligence Group, ARMORY CASTLE exploits BeyondTrust, 15 Oct 2026	cloud.google.com/blog/topics/threat-intelligence/armory-castle-exploits-beyondtrust	2026-11-03	Confirmed
83	HHS HC3, threat briefing on Chinese state-sponsored cyber activity, 19 Nov 2020 (AHA-hosted copy)	aha.org/system/files/media/file/2020/11/hc3-threat-briefing...	2026-11-03	Confirmed
84	FBI and CISA, PSA on PRC targeting of COVID-19 research, May 2020 (AHA-hosted copy)	aha.org/system/files/media/file/2020/05/Joint_FBI-CISA_PSA...	2026-11-03	Confirmed
85	HHS HC3, China's Five-Year Plans and the HPH, 6 May 2021 (AHA-hosted copy)	aha.org/system/files/media/file/2021/05/HC3_Threat_Brief...	2026-11-03	Confirmed
86	HHS HC3, Threat Profile on China-based threat actors, 16 Aug 2023 (AHA-hosted copy)	aha.org/system/files/media/file/2023/08/Hc3-tlp-clear-threat-profile...	2026-11-03	Confirmed
87	US DOJ, APT40 indictment (S.D. Cal., returned 28 May 2021)	justice.gov/opa/press-release/file/1412916/download	2026-11-03	Confirmed
88	FBI, APT 40 cyber espionage activities wanted notice	fbi.gov/wanted/cyber/apt-40-cyber-espionage-activities	2026-11-03	Confirmed
89	FBI, Fujie Wang wanted notice	fbi.gov/wanted/cyber/fujie-wang	2026-11-03	Confirmed
90	FBI, APT 10 Group wanted notice	fbi.gov/wanted/cyber/apt-10-group	2026-11-03	Confirmed
91	The Record, Chinese national arrested in Italy over HAFNIUM and COVID research hacking	therecord.media/chinese-national-arrested-italy-hafnium-covid	2026-11-03	Reported
92	The Record, Chinese hacker extradited from Italy	therecord.media/chinese-hacker-italy-extradited	2026-11-03	Reported
93	NSA, CISA and FBI, AA21-200B, Chinese state-sponsored cyber operations (archived)	cisa.gov/news-events/cybersecurity-advisories/aa21-200b	2026-11-03	Confirmed
94	US DOJ, domain seizure affidavit (QScan, QTRouter and QTFY actors), Aug 2026	justice.gov/d9/2026-08/domain_seizures_affidavit_0.pdf	2026-11-03	Confirmed
95	CNN via CP24, US says Chinese hackers hit hospitals, NASA, Senate and more, 26 Aug 2026	cp24.com/news/world/2026/08/26/...	2026-11-03	Reported
96	FBI, Aquatic Panda cyber threat actors wanted notice	fbi.gov/wanted/cyber/aquatic-panda-cyber-threat-actors	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
97	NSA, CISA and FBI, AA22-158A, PRC exploitation of telecoms and network devices, 7 Jun 2022	cisa.gov/news-events/cybersecurity-advisories/aa22-158a	2026-11-03	Confirmed
98	CISA and partners, Identifying and Mitigating Living Off the Land Techniques, 7 Feb 2024	cisa.gov/resources-tools/resources/identifying-and-mitigating-living-land-techniques	2026-11-03	Confirmed
99	NSA, CISA and FBI, AA22-279A, top CVEs exploited by PRC actors, 6 Oct 2022	cisa.gov/news-events/cybersecurity-advisories/aa22-279a	2026-11-03	Confirmed
100	CISA, Known Exploited Vulnerabilities catalog, JSON feed (version 2026.11.03)	cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json	2026-11-03	Confirmed
101	TeamStarWolf library, ATT&CK v20.0 technique data with tactics and CAPEC mappings (`technique_profiles.jsonl`)	github.com/TeamStarWolf/TeamStarWolf/.../data/attack/technique_profiles.jsonl	2026-11-03	Confirmed

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Related references

- Threat Actors, Malware Families, Notable Incidents, CVE Reference
- Other case studies: The SharePoint ToolShell Mass Exploitation, Log4Shell (CVE-2021-44228), SolarWinds SUNBURST, Stuxnet; companion studies: APT41: The Public Record, Operation Exchange Marauder, Change Healthcare 2024, Sandworm Operations
- Groups: G0006 APT1, G0045 menuPass, G0065 Leviathan, G0128 ZIRCONIUM, G0027 Threat Group-3390, G0125 HAFNIUM, G0096 APT41, G1017 Volt Typhoon, G1045 Salt Typhoon, G1048 UNC3886, G0004 Ke3chang, G0022 APT3, G0009 Deep Panda, G0143 Aquatic Panda, G1054 MirrorFace, G0129 Mustang Panda, G0114 Chimera, G1023 APT5, G0011 PittyTiger, G0026 APT18
- Campaigns: C0017, C0040, C0035, C0039, C0049, C0047, C0052, C0056, C0058, C0060, C0029, C0012
- Software: S0013 PlugX, S0596 ShadowPad, S0154 Cobalt Strike, S0002 Mimikatz, S0020 China Chopper, S0032 gh0st RAT, S1206 JumbledPath, S1144 FRP, S1154 VersaMem, S9015 BRICKSTORM
- Techniques: T1190 Exploit Public-Facing Application, T1133 External Remote Services, T1584.008 Network Devices, T1505.003 Web Shell, T1003.003 NTDS, T1090.003 Multi-hop Proxy, T1059.008 Network Device CLI, T1602.002 Network Device Configuration Dump, T1685.005 Clear Windows Event Logs

- D3FEND: Software Update, Network Traffic Community Deviation, Credential Hardening, Multi-factor Authentication
- Navigator layers, including this study's layer, threat-actors-china.json